The Signaling Gap:

State-Owned Media and the Strategic Management of Sentiment during the Russian Invasion of Ukraine

Donald Moratz and Sarah Gerstein

June 2, 2026

Abstract

Alliance commitments are designed to facilitate cooperation, but they can also prevent states from expressing disagreement through formal diplomatic channels, even when they have reason to dissent. We argue that state-owned media provides an institutionally mediated channel for constrained states to signal distance from a patron's position to foreign governments, occupying a middle ground between formal channels that are too visible and private diplomacy that lacks credibility. Analyzing state media coverage before and after the 2022 invasion of Ukraine, we find that the invasion produced a larger negative shift in sentiment toward Russia in states with Russian security agreements than in neutral states. This occurred despite these states refusing to formally condemn the invasion. Our analysis bridges a gap between political science, which has largely treated state media as a domestic tool, and Western intelligence services, which have monitored it as a window into regime preferences since the Cold War.

Introduction

States that depend on a powerful patron for their security do not always have the luxury of saying what they think. Formal diplomatic channels—UN votes, official statements, public breaks—become liabilities when using them risks retaliation from the very power that guarantees a state's survival. Yet when a patron's actions draw international condemnation, silence risks being seen as endorsement. This bind is not unique to international relations. Hirschman (1970) established that when exit is too costly and direct voice risks retaliation, actors seek indirect channels to communicate dissatisfaction. Research on organizational dissent has documented the same logic in hierarchical institutions: when internal voice is foreclosed, subordinates route disagreement to external audiences who have leverage over the principal (Kassing, 1997). In international politics, the problem is sharpest. States bound by security alliances to a patron that has taken internationally condemned actions cannot publicly break with that patron without risking the very security guarantee they depend on. The question is what channels remain available, and when are they credible?

We argue that state-owned media provides such a channel. State media is a formal institution under government control, but it is not directly attributable as a state position in the way a diplomatic communiqué or UN vote is. This makes it usable precisely when directly attributable channels are too costly. Unlike private diplomatic channels, which external audiences cannot verify and can dismiss as cheap talk (Fearon, 1997), state media is publicly observable—giving the signal reach that private communication lacks. The signal is readable because external observers—particularly Western intelligence services—understand state media to reflect regime preferences. This is not a speculative claim. Western governments have monitored state-owned media as a source of intelligence about regime preferences since the Cold War (BBC Monitoring, 2016; Cohen, 1987; Favere, 2024), treating shifts in tone and emphasis as indicators of policy direction within regimes whose formal diplomatic output is tightly controlled (Williams & Blum, 2018). And the signal is costly: allowing negative coverage of a security patron risks provoking a response, which makes it informative about the sender's alignment. We show that constrained states exploit this channel to signal distance from a patron's position to foreign audiences.

Despite this well-established monitoring infrastructure, political science has not formalized the international signaling function of state media. The discipline has produced a substantial literature on how governments use media to manage domestic audiences—setting agendas (McCombs, 2002), shaping attitudes in weak democracies (Enikolopov et al., 2011), and managing information to maintain power (Guriev & Treisman, 2022). But this work is oriented almost entirely toward domestic publics. International relations scholarship, meanwhile, has focused on formal, directly attributable instruments—UN votes, official statements, military deployments—as the channels through which states communicate resolve and commitment (Fearon, 1997; Fuhrmann & Sechser, 2014). The result is a gap: the same outlets that political science studies as tools of domestic control also produce content that intelligence services treat as international signal, yet the discipline has not developed a theoretical framework that accounts for both functions simultaneously. Our analysis addresses this gap. We demonstrate that the signals practitioners monitor are theoretically structured and empirically identifiable—they follow directly from the logic of alliance constraints, and they discriminate between competing theoretical accounts.

The Russian invasion of Ukraine on February 24, 2022—the largest land war in Europe since World War II—confronted Russia's formal security partners with a bind between their patron and the West. Formal condemnation risked retaliation from the patron that guarantees their security; silence risked being categorized alongside an internationally condemned aggressor. States bound to Russia through the Collective Security Treaty Organization and other formal security agreements had genuine reasons to oppose the invasion. It violated the sovereignty norms that protect small states and threatened their substantial economic ties with the West. But open condemnation of a security patron that had already demonstrated its willingness to use force in its near abroad was not a viable option. The invasion thus provides an unusually clear test of our argument: a discontinuity in the information environment, a well-defined alliance structure that constrains formal channels of dissent, and competing theoretical predictions that the data can adjudicate.

We test this argument by exploiting variation in formal security relationships with Russia across twelve states in Russia's near abroad, drawing on a corpus of several hundred thousand newspaper articles published before and after the invasion.[1] Four states were bound to Russia through the Collective Security Treaty Organization or equivalent bilateral security agreements at the time of the invasion. Four occupied broadly neutral positions, maintaining distance from both Russian security structures and full Western integration. The remaining states are NATO members. We employ a generative AI model (GPT-4o) to measure sentiment toward Russia at the article level, validating performance against a human-coded gold standard. Our identification strategy uses the discontinuity of February 24 to estimate the causal effect of the invasion on media sentiment within each group through a regression discontinuity in time design. To compare across groups, we leverage a Regression Difference-in-Discontinuity Design (RDID). The discontinuity of the invasion substantially reduces confounding concerns, though comparisons across groups remain correlational in nature. We subject these comparisons to extensive robustness checks, which we detail in our methods section.

Our findings build sequentially. We first establish that the invasion produced a significant negative shock to media sentiment toward Russia across the region—without this baseline, the divergence we identify in subsequent tests would be uninterpretable. We then show that this decline was not uniform: independent media responded significantly more negatively than state-owned media, consistent with strategic management of coverage by state outlets. The critical test of our argument comes in the comparison across security relationships. A straightforward domestic audience management account would predict that Russia-aligned states suppress negative coverage to protect the alliance relationship. Instead, we find the opposite: state-owned media in states with Russian security agreements shifted more negatively toward Russia than state-owned media in neutral states, despite these same states refusing to condemn the invasion through formal diplomatic channels. This pattern is consistent with constrained states using media to signal distance from their patron precisely because formal channels were

[1] The twelve states span three categories of alignment. Russia-aligned: Armenia, Azerbaijan, Belarus, and Kazakhstan. Neutral: Georgia, Moldova, Kosovo, and Serbia. NATO-aligned: Albania, Hungary, and Turkey. Ukraine is included in the dataset but excluded from the main analysis given its direct role as the invaded party.

unavailable.

This paper makes two primary contributions. First, our analysis demonstrates that the signals practitioners have long monitored are theoretically structured and empirically identifiable. The patterns we document are not idiosyncratic to a single crisis but follow directly from the logic of alliance constraints: states whose formal channels are constrained by dependence on a powerful patron produce systematically different media signals than states whose channels remain open. This suggests that state media sentiment is not merely noise but a legible instrument of international communication, one that the discipline can study with the same rigor it applies to formal diplomatic behavior. Second, we identify an understudied mechanism of institutionally mediated signaling under alliance constraints. The signaling literature has focused on how states communicate resolve to adversaries or commitment to allies through formal channels; we demonstrate that when those channels are foreclosed, states turn to institutionally mediated ones, and that the resulting signals follow predictable patterns rooted in the structure of their security commitments. Methodologically, our approach offers two contributions: the pairing of prompted large language model classification with a regression discontinuity design provides a template for large-scale analysis of media signals, and our combination of coarsened exact matching with the sensitivity framework of Cinelli and Hazlett (2020) provides a transparent procedure for assessing the robustness of cross-group comparisons when group membership is observational.

The article proceeds as follows. We first develop our theoretical argument and derive three hypotheses about media divergence under alliance constraints. We then present the case, justifying our sample and identification strategy. The following sections describe our data and research design before presenting results. We conclude by discussing implications for how states signal preferences under geopolitical constraint.

Theory and Hypotheses

Formal signaling theory establishes that credible diplomatic signals require the signaling state to incur costs that make bluffing irrational (Fearon, 1997). The logic is straightforward—if a signal is costless to send, it carries no information, because any state could send it regardless of its true intentions. The literature has identified two primary mechanisms through which states generate these costs. The first is audience costs, whereby leaders stake domestic political capital on a public position, making reversal politically expensive (Fearon, 1997). The second involves sinking material costs through actions such as troop deployments, arms transfers, or economic sanctions that are costly to undertake regardless of the outcome (Blankenship & Lin-Greenberg, 2022; Fuhrmann & Sechser, 2014). Both mechanisms share a common feature—they operate through formal, visible, and directly attributable channels. UN votes, official diplomatic statements, military deployments, and public breaks with allies are the instruments this literature has overwhelmingly focused on, precisely because their visibility is what generates the costs that make them credible.

This focus leaves less to say about what happens when those channels are foreclosed. Alliance commitments can create exactly this condition. Alliances do not merely aggregate capabilities against external threats; they also constrain their members (Weitsman, 2004). States that tether themselves to a powerful patron gain security, but they do so by trading autonomy (Morrow, 1991), and one dimension of that lost autonomy is the freedom to publicly dissent. Formal condemnation of a security patron risks what Snyder (1997) identifies as the central dilemma of alliance politics, abandonment. A CSTO member that votes against Russia at the United Nations is not merely casting a costly vote; it is signaling a willingness to break with the state that underwrites its security, an action that could trigger the withdrawal of that guarantee entirely. The result is that formal channels lock precisely when states have the strongest reason to communicate dissent.

What do constrained states do when formal channels are unavailable? The signaling literature has had less to say about this question, in large part because its theoretical toolkit was built to analyze the instruments where costs are most visible. We argue that state-owned media provides one answer.

State-owned media is not understudied. The discipline has produced a substantial literature on how governments use media to manage domestic audiences. News media sets agendas by determining which issues receive public attention (McCombs, 2002) and primes audiences by shaping the criteria they use to evaluate leaders (Iyengar & Kinder, 2010). The framing of an issue shapes how audiences form opinions about it (Chong & Druckman, 2007), and careful framing can have a dramatic impact on public support for government actions, particularly during crises (Gershkoff & Kushner, 2005). Political parties exert significant influence over their followers' opinions and can leverage this to reshape support for policies (Slothuus & Bisgaard, 2021). Work on autocratic media management has extended these insights to non-democracies. Modern autocracies maintain power through sophisticated information control rather than outright censorship (Guriev & Treisman, 2022). The underlying resource wealth of a state shapes the incentives governments face in managing media freedom (Egorov et al., 2009), and state media can powerfully shape attitudes in countries with weak democratic institutions (Enikolopov et al., 2011). Within the post-Soviet space specifically, Oates (2006) argues that state media serves most directly as a tool for elites to engage in agenda setting, highlighting its effectiveness in managing political discourse. This literature has been productive, and much of it applies directly to the states in our sample. But its orientation is almost entirely toward domestic audiences—how governments shape what their own publics think. The possibility that the same outlets simultaneously communicate to external audiences has received far less attention.

The practitioners tasked with understanding regime preferences have operated on a different assumption. Western intelligence services began systematically monitoring state-owned media during World War II, with the BBC Monitoring Service established in 1939 and the U.S. Foreign Broadcast Information Service in 1941 (Favere, 2024). During the Cold War, this practice became its own analytical tradition. Western analysts read Soviet and satellite state media not as journalism but as political text, treating shifts in tone, emphasis, and attribution as indicators of factional dynamics and policy direction within regimes whose formal diplomatic output was tightly controlled (Cohen, 1987). Cohen (1987) argues that diplomatic signaling through media is ambiguous by design—it allows governments to communicate positions while preserving deniability, a feature well understood by the intended recipients. This dynamic has only intensified. Open-source intelligence has become formalized within diplomatic and intelligence institutions (Manor, 2019; Williams & Blum, 2018), and the U.S. Intelligence Community has elevated state media monitoring to a core component of its analytic strategy (USIC, 2024). The British Parliament's Intelligence and Security Committee described BBC Monitoring as "indispensable" to the national security apparatus (BBC Monitoring, 2016). Nor is this monitoring exclusively Western; Flamer (2023) documents how Hamas systematically monitored Israeli media to extract signals about Israeli policy intentions, suggesting that the practice of reading state-adjacent media as political signal is common knowledge across a range of actors. The institutional infrastructure is not incidental to our argument—it is part of the mechanism. States in our sample operate in an environment where they can reasonably expect that shifts in their state media coverage will be observed, interpreted, and acted upon.

The domestic and international functions of state media are not separate phenomena. The same outlet that manages domestic opinion about a security patron also produces content that foreign analysts monitor for signals of regime preferences. The theoretical tools to understand each function exist in the literature—agenda setting, framing, and autocratic media management on one side; diplomatic signaling, credibility, and monitoring infrastructure on the other—but they have not been assembled into a framework that accounts for both simultaneously. Our argument connects them. We do not claim that all state media operates as an international signaling channel. The argument is sharpest for state media in states where government control over editorial content is well established and widely understood by external observers—conditions that characterize the post-Soviet and nearby states in our sample but that do not describe, for instance, public broadcasters in consolidated democracies. It is precisely because foreign analysts take it as given that outlets like Kazakh or Belarusian state media reflect regime preferences that deviations in coverage become informative.

If state media is a strategic tool, its use as a signaling channel requires that the signal be readable as a deliberate choice and costly enough to be credible. Gehlbach and Sonin (2014) provides a foundation for readability. In their model, governments use state-owned media to advance their agenda, producing systematically higher bias in state outlets than in independent media. This establishes the baseline expectation of alignment from which deviation becomes readable—external observers can distinguish a regime choice from journalistic noise. Readability is what makes state media, rather than independent media, the relevant signaling channel. A regime that simply loosened constraints on independent coverage would produce a shift in tone, but external observers would have no basis for attributing that shift to the regime rather than to editorial independence. State media is readable as a regime choice precisely because external observers understand it to be under government control. States in this context are not merely following elite cues or responding to media pressure on foreign policy (Robinson, 2002); they are deploying a controlled outlet as an instrument of international communication. There is evidence that state media already operates this way in the region. Peisakhin and Rozenas (2018) show that exposure to Russian state television increased support for pro-Russian candidates in Ukraine's 2014 elections, demonstrating that state media shapes political attitudes across borders, not only within them.

Readability alone is not sufficient. The signal must also be costly to send. Allowing negative coverage of a security patron risks provoking a response—and a state with no reason to signal distance from the patron would not accept that risk. The costliness of the signal is what makes it informative: it shifts the receiver's beliefs about the sender's alignment. The signal's value to the sender does not require that the receiver learn the sender's "true" preferences with certainty. It requires only that, after observing the shift in state media coverage, foreign analysts assign a higher probability to the interpretation that the client state is not fully aligned with the patron than they did before. This is the standard signaling logic of Fearon (1997) applied to an understudied channel: the signal is costly (allowing negative coverage risks patron retaliation), therefore informative (a state fully captured by the patron would not accept that risk), and therefore belief-shifting.

The strategic calculus facing constrained states can be stated explicitly. The cost of signaling through state media is the risk of patron retaliation. Russia's interventions in Georgia in 2008 (Allison, 2008) and Ukraine in 2014 demonstrate that Moscow is willing to punish states that break formally with its security framework—and the possibility that media-level dissent could be interpreted as the beginning of such a break is the source of cost. The benefit of signaling is shifting Western beliefs about alignment, which may affect sanctions treatment, diplomatic engagement, and the preservation of economic relationships with the West—relationships that, as we detail in the Case section, are substantial for the states in our sample. The cost of *not* signaling is being categorized alongside the patron by Western governments, risking the same diplomatic isolation and economic penalties despite having reasons to oppose the patron's actions. It is the asymmetry between these costs—the bounded risk of media-level dissent versus the potentially large cost of being lumped in with an internationally condemned patron—that makes signaling through state media a rational strategy.

The core logic—readability and costliness producing a credible, belief-shifting signal—does not require that the channel be usable repeatedly. Even a single signal is valuable if it shifts the receiver's beliefs. In practice, however, states face iterated interactions with their patrons, and whether the channel remains available over time depends on additional features of the relationship. Two features support sustainability in this setting. First, state-owned media operates below the threshold of a formal diplomatic break. Russia has demonstrated willingness to punish formal challenges to its security framework—Georgia's NATO aspiration and Ukraine's EU association agreement both preceded military intervention. But state media coverage, while observable, does not constitute a formal diplomatic act. The institutional separation between the government and state-owned media outlets provides what Entman (2004) describes as a cascade structure: frames originate with the executive and filter through institutional layers to publication, and these layers—however nominal—provide structural separation. A challenged government can disavow unfavorable coverage as editorial judgment rather than state policy, preserving the patron relationship even as the signal reaches its intended audience. The signal is thus costly enough to be credible but institutionally buffered enough to avoid crossing the threshold that has historically triggered patron punishment.

Second, mutual dependence raises the patron's cost of responding even to signals it detects. Russia has invested substantial diplomatic and military capital in maintaining the Collective Security Treaty Organization as a counterweight to NATO expansion (Allison, 2008), and the states in our sample serve as important economic and strategic partners. Punishing a client state for media-level dissent risks pushing that state toward the West—precisely the outcome Russia's alliance framework is designed to prevent. The sustainability of the signaling channel thus depends in part on the structure of the patron-client relationship: where both parties are mutually dependent, the patron's cost of punishing media-level signals is high enough that the channel remains viable.

A natural concern is that observed divergence between state media and official positions reflects internal incoherence rather than strategic communication. Weak democracies and authoritarian regimes are not monolithic. Authoritarians face multiple factions they must balance (Svolik, 2012), and the elites who surround a ruler are not merely extensions of the leader but actors with their own preferences and survival calculations (Bueno de Mesquita, 2003). Dictators manage these competing interests through institutional bargaining and cooptation (Gandhi, 2008), and state-owned media outlets, as institutions within the regime, may reflect these competing internal voices rather than a single coordinated signal. Expressions of dissent even in the most constrained autocracies are consistent with factional interests rather than top-down direction (Malesky & Schuler, 2010). Our data does not allow us to distinguish between these two mechanisms. But whether state media diverges from formal positions because of strategic choice by leadership or because of elite factional pressure, the observable result is the same: a costly, informative signal of distance from the patron's position. The receiver's belief update—the increased probability that the client state is not fully aligned with the patron—is the same regardless of which internal process generated it. The divergence itself is analytically meaningful, and our regression discontinuity design identifies the causal effect of the invasion on state media sentiment regardless of which internal mechanism produced it. We develop this identification strategy in our methods section.

Our theoretical argument requires that the invasion of Ukraine represented a genuine negative shock to media sentiment across the region—without this baseline, the divergence we identify in H2 and H3 would be uninterpretable. We first establish this empirically:

Hypothesis 1: The invasion of Ukraine produced a significant decline in media sentiment toward Russia across both state-owned and independent outlets in our sample.

With this baseline established, we turn to the divergence between state-owned and independent media. State-owned media in our sample operates under political constraints that independent media does not face. We therefore expect that state-owned media will manage coverage more carefully than independent media, producing a smaller decline in sentiment toward Russia after the invasion. This test is also informative about readability: if state outlets operate under political constraints that shape their coverage, we should observe a measurably different response to the same shock—and it is precisely this differential that makes state media coverage interpretable as a regime choice rather than independent journalism.

Hypothesis 2: State-owned media will report negative sentiment towards Russia after the invasion to a lesser extent than independent media.

A straightforward domestic audience management account would predict that states with formal security agreements with Russia suppress critical coverage of the invasion—protecting the alliance relationship means keeping the domestic public from reading negative coverage of Russia. Our signaling argument predicts the opposite: precisely because formal diplomatic condemnation is too costly, these states use state-owned media as an institutionally mediated channel to signal distance to external audiences, producing more negative coverage than neutral states whose formal channels remain open. The comparison to neutral states is critical. States without formal security agreements with Russia face no alliance constraint on their formal channels and therefore have no signaling need that state media must fill. If anything, neutral states in Russia's near abroad have a mild incentive to avoid antagonizing Moscow, which should keep state media coverage relatively flat. If state media in Russia-aligned states nonetheless shifts more negatively than in neutral states, the pattern cannot be explained by a mechanical response to the invasion—it requires the kind of strategic logic our argument describes. That the domestic account and the signaling account generate opposite predictions for H3 provides additional leverage for distinguishing between them.

Hypothesis 3: State-owned media in states with formal security agreements with Russia will respond more negatively to the invasion than state-owned media in neutral states.

If domestic audience management were the primary driver, we would further expect coverage patterns to track the size of ethnic Russian populations across our sample—since Russia explicitly justified the invasion as protection of ethnic Russians abroad, regimes with larger ethnic Russian minorities would face the strongest domestic incentive to suppress critical coverage. We return to the opposite-predictions test and the ethnic Russian robustness check in our discussion.

The Case

The February 2022 invasion of Ukraine was a direct violation of the UN Charter's prohibition on the use of force against the territorial integrity of a sovereign state (Nations, 1945)—a norm that Russia, as a permanent member of the Security Council, had formally committed to uphold. The international community largely condemned the attack. The Council of Europe suspended Russian participation, and the UN Security Council took up a draft resolution condemning the invasion, which Russia vetoed (Walker, 2024). Condemnation was the majority response but not a universal one, and the states that withheld it did so for varied economic and geopolitical reasons. The cases at the center of our analysis faced a more specific constraint: bound to the aggressor through formal security agreements, public condemnation meant breaking with the very patron on which their security depended.

The states in our sample fall into three broad categories of alignment, each of which shaped how they could respond. Armenia, Belarus, and Kazakhstan are all members of the Collective Security Treaty Organization (CSTO), Russia's primary multilateral security alliance in the post-Soviet space. Azerbaijan, while not a CSTO member, signed a Declaration on Allied Interaction with Russia in February 2022, two days prior to the invasion (Huseynov, 2022), placing it in a similar position of formal alignment. CSTO membership creates institutional pressure toward solidarity, making public condemnation of the invasion a direct challenge to the alliance framework itself. At the same time, all four states maintained substantial trade relationships with Western nations, giving them a significant economic stake in opposing the invasion even as their security commitments constrained them from doing so[2].

The second category comprises states we treat as broadly neutral: Georgia, Moldova, Kosovo, and Serbia. Each has maintained distance from both Russian security structures and full Western integration, though for different reasons and with different relationships to Russia's sphere of influence. Georgia and Moldova have both faced Russian territorial incursions in recent years (Cheterian, 2009; Potter, 2022), while Serbia and Kosovo's distance from both Russia and Western integration reflects different dynamics entirely. Serbia has traditional ties to Moscow, and the tension between those ties and its EU candidacy has shaped its stance on the war (Radeljić & Özşahin, 2023). Kosovo, meanwhile, is dependent on Western recognition for its own sovereignty claim, though it sits outside formal security structures on both sides (Newman & Visoka, 2016). While our neutral states reflect a broad mix of relationships both with the West and with Russia, this mix allows them to function as an appropriate baseline of comparison for states with explicit security agreements.

The third category comprises states aligned with Western institutions: Albania and

[2] The level of trade reliance expressed as a percent of total trade (imports plus exports) originating from the west (Europe and North America) for each country in 2021 was Belarus: 22%, Armenia: 34%, Kazakhstan: 46%, and Azerbaijan: 72% (Bank, 2021). These trade relationships persisted through the invasion period; Western trade shares for Kazakhstan, Azerbaijan, and Armenia in 2022 were 48%, 73%, and 29% respectively, comparable to 2021 levels.

Hungary are both full NATO members, and Ukraine, while not a NATO member at the time of the invasion, had formally declared NATO membership a strategic objective. Turkey presents a more complex case: formally a NATO member since 1952, Ankara declined to join Western sanctions against Russia and instead positioned itself as a mediator between the two parties, reflecting the increasingly independent foreign policy orientation it had pursued since normalizing relations with Moscow after 2016 (Mankoff, 2022). We treat Turkey as NATO-aligned for the purposes of our sample construction, while acknowledging that its behavior during the invasion did not fully mirror that of other alliance members. Ukraine is included in our dataset but excluded from the main analysis given its direct role as the invaded party.

The constraints facing Russia-aligned states in February 2022 did not emerge overnight. Russia's willingness to use force in its near abroad had been established clearly by 2014, when Russia annexed Crimea following political upheaval in Ukraine and backed separatist forces in the Donbas. The Minsk Agreements, brokered by Western nations, called for a ceasefire and reaffirmation of Ukrainian sovereignty, but violence continued to escalate through the intervening years (Walker, 2023). By November 2021, Russia had mobilized troops along the Ukrainian border, and intelligence from both the U.S. and U.K. was indicating that a full-scale invasion was imminent (Walker, 2023). Russia simultaneously demanded that NATO commit to admitting no new members—explicitly naming Ukraine and Georgia—signaling that states in its near abroad would face consequences for Western alignment (Jazeera, 2022). For CSTO members and Russia-aligned states, this context made formal diplomatic condemnation of the invasion not just politically costly but institutionally constrained.

February 24 nonetheless represents a discrete and dramatic rupture in the information environment. While the troop buildup had proceeded for months, the full-scale invasion—with missile strikes on Kyiv and ground forces crossing the border simultaneously—was of a different order than the slow escalation that preceded it. Russia had conducted large-scale military exercises with Belarus near the Ukrainian border on February 10, with Western intelligence reporting up to 140,000 troops positioned at the border (Walker, 2024). Even so, the decision to launch a comprehensive invasion rather than a more limited operation represented a threshold that most observers had not fully anticipated. Although the United States and United Kingdom had taken the unusual step of publicly disclosing intelligence warnings of a full-scale assault in the preceding weeks (Dylan & Maguire, 2022), a large-scale invasion was widely regarded as improbable—even among senior Western officials with access to that intelligence (Banco et al., 2023)—and many governments and analysts, including within the states in our sample, treated a limited operation in the Donbas as the more likely outcome, such that the eventual scale registered as a strategic shock (Götz & Staun, 2022). This discontinuity is the basis for our identification strategy: we treat February 24 as a break in the media environment and use the variation in coverage before and after it to identify the causal effect of the invasion on state media sentiment. We describe our data and measurement approach in the following section.

The Data

Our analysis draws on the High-Quality Media from Aid Receiving Countries (HQ-MARC) corpus, a collection of news articles from prominent domestic media outlets built through human-supervised, source-specific web scraping (Moratz et al., 2025). HQMARC's design prioritizes comprehensive capture from each source, enabling measurement of both article-level content and source-level patterns of coverage. Articles published in non-English languages were translated using neural machine translation models validated against human translations (Moratz et al., 2025). For the present study, we collected all Russia-related articles published between August 1st, 2021 and August 30th, 2022 from 12 countries and 70 sources across the region. Russia-related articles were identified using a curated keyword list of over 500 terms associated with Russia, including prominent politicians, government institutions, and companies; the full keyword list is documented in Moratz et al. (2025). The 12 countries in our sample span three categories of alignment with Russia, which we describe in detail in the Case section. We omit Ukraine from the analysis given its direct role as the invaded party, leaving 11 countries, 61 sources, and 174,094 articles.

A central distinction in our analysis is between state-owned and independent media. We classify an outlet as state-owned if it is either directly owned and operated by the state or owned and operated by close affiliates of the executive branch, such as family members or cabinet ministers. Ownership was traced through a detailed desk review of media ownership reports, press freedom assessments, and consultations with regional experts, following the procedure described in Moratz et al. (2025). This classification yields 17 state-owned and 44 independent sources across the 11 countries in our sample, with at least one state-owned outlet in each country. The distinction is theoretically central to our argument: external observers—particularly Western intelligence analysts—understand these outlets to be under government control, which is what makes shifts in their coverage readable as regime choices rather than independent editorial judgment (Gehlbach & Sonin, 2014). Figure 1 presents the daily volume of Russia-related articles by media type over the sample period. The sharp spike in coverage at the invasion is visible across both state-owned and independent outlets, though independent media produces substantially more Russia-related content throughout—a difference addressed by the source-level standardization described in the Research Design.

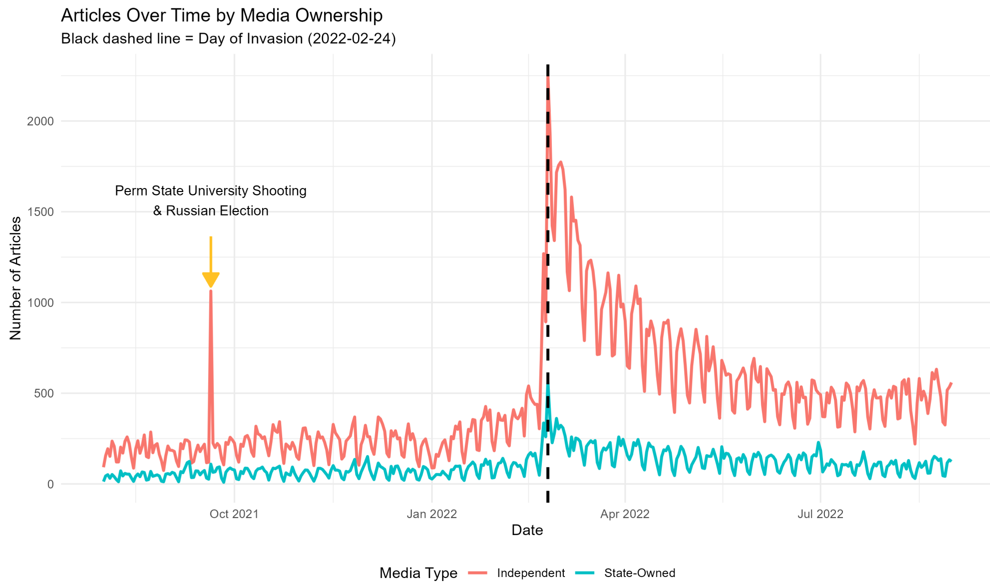

Figure 1: Daily count of Russia-related articles by media ownership type. The black dashed line marks February 24th, 2022.

We coded each article along two dimensions using a zero-shot GPT-4o Turbo classifier applied to the article title and first two sentences: article topic and sentiment toward Russia.[3] Our use of a prompted large language model rather than an off-the-shelf sentiment classifier reflects a measurement choice. Standard transformer-based sentiment models (e.g., BERT-based classifiers) capture the overall tone of an article rather than sentiment toward a specific actor. In our setting, this distinction is critical: an article reporting on the invasion could be coded as negative in tone regardless of whether it criticizes Russia, reports neutrally on the conflict, or frames Russia's actions sympathetically. Our analysis requires sentiment toward Russia specifically—the dimension along which state-owned and independent media are theorized to diverge. The zero-shot GPT-4o classifier allows us to specify this target directly in the prompt, measuring the article's orientation toward Russia as an actor rather than the valence

[3]The sentiment and topic classification tasks are substantially simpler than the 20-category civic space classification in Moratz et al. (2025), and zero-shot performance was high enough (F1 of 0.92 for sentiment) that fine-tuning was unnecessary. Applying the classifier to the full article text did not produce significant performance gains, so we used a truncated portion to reduce computational costs. This choice is not merely expedient: Moratz et al. (2025) finds that classification on the title and first two sentences achieves the best performance against human coding on the underlying corpus. Because our gold standard here was likewise coded on this same truncated input, the validation results reflect the classifier's performance on exactly the text it processes rather than on idealized full-article input.

of the events described. This orientation is a property of the coverage itself—how favorably or unfavorably an article portrays Russia—and not a measure of the outlet's editorial stance: a state outlet that selects or foregrounds coverage unfavorable to Russia registers as a negative shift whether or not it editorializes in its own voice, which is precisely the channel through which our account expects the signal to operate. Topics were classified as one of five categories: political, economic, military, cultural, or other. Sentiment toward Russia—encompassing the Russian state, military, government ministers, and Vladimir Putin—was coded on a ternary scale: -1 (negative), 0 (neutral), or 1 (positive). We also classify whether Russia is a main actor in each article; this measure is used in supplementary analyses reported in the appendix.

To validate the classifier, we constructed a gold standard of 100 randomly selected articles, adversarially double-coded by the coauthors and debated to consensus. The F1 score for the sentiment classifier is 0.92 and for the topic classifier is 0.80, with confusion matrices reported in Figures A1 and A2. The high F1 scores provide confidence that the gold standard, though modest in size, is adequate for validation. One feature of the classifier merits note: it is more likely than human coders to classify political articles as other, an artifact of the overlap between these categories in the data.[4] Representative examples of articles coded at each sentiment level are presented in the appendix.

Research Design

We leverage two complementary designs to test our hypotheses. First, we use a regression discontinuity design (RDD) to estimate the causal effect of the invasion on media sentiment toward Russia within groups of outlets (Cattaneo & Titiunik, 2022; Cattaneo et al., 2019). Second, we use a regression difference-in-discontinuities (RDID) design to estimate whether the invasion produced *differential* effects on sentiment across groups (Grembi et al., 2016; Tramontin Shinoki et al., 2024). The RDD identifies within-group

[4] A manual review of 100 randomly sampled articles coded as "other" reveals that the category comprises crime, accidents, and disasters (approximately 20%), COVID-19 and health news (15%), bilateral administrative and logistical matters such as flight routes and travel restrictions (15%), human interest and sports (12%), and miscellaneous international news mentioning Russia tangentially (13%). The remaining approximately 20% are articles with political valence—espionage, journalist persecution, refugee flows—that the classifier codes as "other" rather than "political," consistent with the overlap noted above.

effects under standard continuity assumptions. The RDID extends this logic to compare discontinuities across groups—state-owned versus independent media (hypothesis 2) and Russia-aligned versus neutral state media (hypothesis 3)—under a parallel-discontinuities assumption: that, absent the group-specific mechanism, both groups would have experienced the same discontinuity at the cutoff. We are transparent about this hierarchy of credibility throughout the analysis: within-group discontinuities are causal, while cross-group comparisons leverage the parallel-discontinuities assumption and are subjected to extensive robustness checks.

The Regression Discontinuity Design

We exploit the discontinuity of the Russian invasion on February 24$^{\text{th}}$, 2022 to estimate the causal effect of the invasion on sentiment toward Russia in news coverage across our sample. The forcing variable is the date-time distance between an article's publication and the date of invasion.[5] Treatment is assigned deterministically: all articles published at or after the invasion are treated. Formally, treatment T is assigned by the distance between publication date-time D_p and the invasion D_I:

$$T = \begin{cases} 1 & \text{if } D_p \geq D_I \\ 0 & \text{if } D_p < D_I \end{cases} \tag{1}$$

The key identifying assumption is that potential outcomes are continuous through the cutoff—that is, absent the invasion, sentiment would have evolved smoothly through February 24$^{\text{th}}$. We believe this assumption is reasonable for two reasons. First, the invasion, while anticipated in the weeks prior, constituted a discrete shock: the decision to invade was made by a single actor (the Russian government) and the precise timing was not known ex ante to the media outlets in our sample. Second, no other event on February 24$^{\text{th}}$ plausibly produced a discontinuity in media sentiment toward Russia

[5] We discuss our use of publication date-time rather than publication date, and our handling of articles with missing timestamps, in the Forcing Variable section.

across the region.[6]

Under continuity, we can identify the local average treatment effect (LATE) of the invasion on sentiment at the cutoff:

$$\widehat{\gamma_{\text{rdd}}} = \lim_{D_p \downarrow D_I} E[Y_p(1) \mid D_p = D_I] - \lim_{D_p \uparrow D_I} E[Y_p(0) \mid D_p = D_I] \tag{2}$$

where $Y_p(1)$ and $Y_p(0)$ represent article-level sentiment toward Russia under the post-invasion and pre-invasion potential outcomes, respectively. For hypothesis 1, this estimand is identified using `rdrobust` (Cattaneo et al., 2019) with the full set of fixed effects (country, source, and topic) and MSE-optimal bandwidth selection (Calonico et al., 2020). We use HC1 heteroskedasticity-robust standard errors clustered at the source level for all RDD specifications.

The Regression Difference-in-Discontinuities Design

Hypotheses 2 and 3 ask whether the invasion produced differential effects across groups. The RDID extends the RDD by comparing the estimated discontinuity in one group to the estimated discontinuity in another (Grembi et al., 2016; Tramontin Shinoki et al., 2024). If state-owned media responded differently from independent media (H2), or if Russia-aligned state media responded differently from neutral state media (H3), the RDID will detect this as a significant difference in the magnitude of the two discontinuities.

No nonparametric framework for the RDID currently exists. We therefore estimate the RDID using OLS within the MSE-optimal bandwidth, applied to a fully interacted specification:

$$Y_p^z = \beta_0 + \beta_1 D_p + \beta_2 T_p + \beta_3 D_p T_p$$

[6] We test this assumption with a placebo cutoff at December 7$^{\text{th}}$, 2021—the date the United States publicly announced that Russia intended to invade—and find no significant discontinuity.

$$+ \beta_4 G_p + \beta_5 T_p G_p + \beta_6 D_p G_p + \beta_7 D_p T_p G_p + \gamma_{\text{topic}} + \varepsilon_p \quad (3)$$

where Y_p^z is the z-scored sentiment of article p (standardized within source; see the Standardization section), D_p is the running variable, T_p is the treatment indicator, G_p is the group indicator (state-owned vs. independent for H2; Russia-aligned vs. neutral for H3), and γ_{topic} is a vector of topic fixed effects. The error term ε_p is clustered at the source level. The RDID estimate is β_5: the differential shift in sentiment at the cutoff for the group of interest relative to the comparison group. We detail the rationale for the z-scored outcome, the bandwidth selection procedure, and inference with few clusters in subsequent sections.

The identifying assumption for the RDID is structurally analogous to the parallel trends assumption in difference-in-differences, applied to discontinuities rather than levels. The specification absorbs level differences between groups (β_4) and differential trends in the running variable (β_6). What remains is the assumption that, absent the group-specific mechanism we theorize, both groups would have experienced the same discontinuity at the cutoff. The threat to identification is a factor that produces a differential *jump* in sentiment at the invasion date, is correlated with group membership, but operates through a channel other than the one our theory describes. We discuss the conditions under which RDID estimates admit a causal interpretation—and why our specification is best understood as a strongly informative comparison rather than a causal claim about group membership—in the Interpreting the RDID section. We detail our sensitivity analysis and robustness checks, which directly address the plausibility of confounding threats, in the Robustness section.

Identification and the Role of Covariates

Isolating the effect of the invasion on sentiment requires careful consideration of the pathways through which the invasion can affect our outcome. The directed acyclic graph in Figure 2 illustrates the core identification challenge. The invasion does not only change *how* outlets write about Russia—it also changes *what* they write about. If military topics spike after the invasion and military coverage is inherently more negative toward Russia, then the estimated discontinuity will conflate genuine shifts in editorial tone with a mechanical recomposition of the article pool. This is a compositional threat to identification, not a confounding variable in the traditional sense: the invasion causes both the topic shift and the sentiment shift, but our quantity of interest is the direct effect on sentiment, not the indirect effect operating through topic composition.

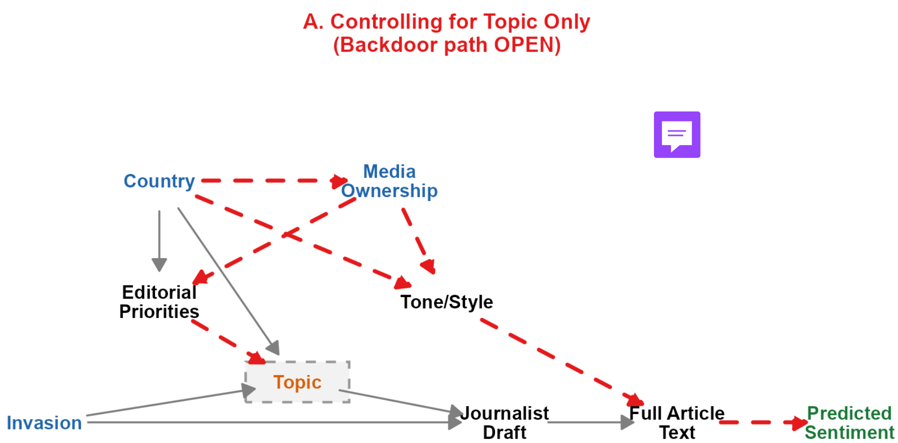

Figure 2: Directed acyclic graph illustrating the identification problem. The invasion affects sentiment both directly and indirectly through changes in topic composition. A naive approach of blocking only on topic introduces a backdoor path through country and ownership.

A naive solution—simply controlling for topic—is insufficient. As Figure 2 shows, topic is a collider on the path between source characteristics (country, ownership) and sentiment. Conditioning on topic alone opens a backdoor path through country and ownership, introducing bias rather than eliminating it. The solution requires blocking all three pathways simultaneously: topic, country, and ownership.

Within our sample, each outlet has a fixed country and ownership type. Source identity therefore subsumes both country and ownership as a finer partition. Controlling for source blocks the backdoor paths through country and ownership while also absorbing source-level differences in topic profiles. Figure 3 illustrates this solution. Topic fixed effects in the regression then absorb remaining within-source variation in topic composition at the cutoff.

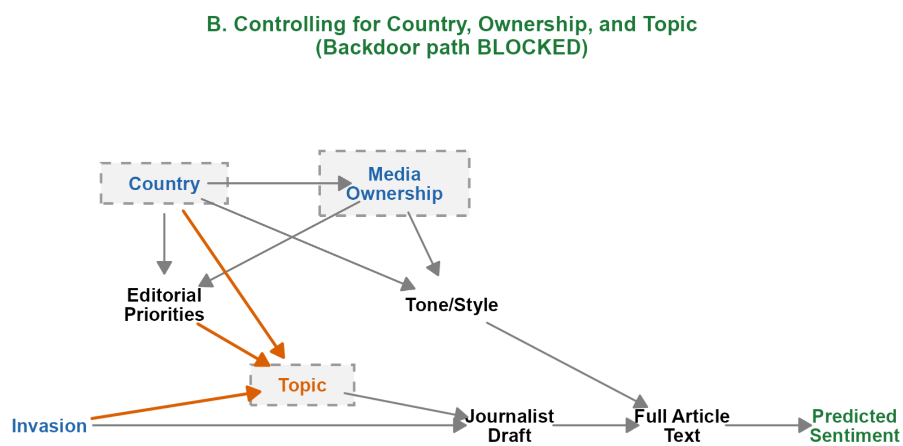

Figure 3: Directed acyclic graph illustrating the identification solution. Controlling for source blocks the backdoor paths through country and ownership. Topic fixed effects absorb compositional shifts within sources.

However, covariate adjustment in the RDD framework is only a valid identification strategy if the covariates being adjusted for do not themselves exhibit a sharp discontinuity at the cutoff (Cattaneo et al., 2023). If they do, the fixed effects absorb part of the treatment effect rather than isolating it. Here, this concern is substantive: the invasion plausibly produces a sharp increase in the share of articles coded as military in nature. If military articles spike discontinuously at February 24$^{\text{th}}$, topic fixed effects may partial out a component of the genuine treatment effect. Relatedly, article volume itself increases sharply after the invasion, changing the composition of the sample at the cutoff. Because manipulation of the forcing variable is not a concern in this setting—outlets do not choose their publication dates in response to the invasion threshold—this volume change reflects a compositional shift rather than a threat to the assignment mechanism, and is addressed by the topic-level checks described below.

We address the compositional concern in two ways. First, we include specifications in which articles coded as military are omitted entirely, removing the compositional channel by construction. Second, we estimate the RDD within each topic separately, allowing us to observe how the discontinuity operates *within* topic categories rather than across them. If the discontinuity is present within non-military topics, the compositional threat cannot account for the main result. Both checks are reported in the Robustness section. Figure 4 shows how the composition of articles changes over time. Military coverage grew steadily in the months preceding the invasion as the troop buildup progressed, and while its share increases further after February 24$^{\text{th}}$, the shift is consistent with the pre-existing trend. The more pronounced compositional change is a reallocation from political to economic coverage in the weeks immediately following the invasion.

A downstream consequence of source-level control is that source identity is collinear with our key partitions of interest—whether an outlet is state-owned or independent (H2) and whether it belongs to a Russia-aligned or neutral state (H3). These group memberships are fixed within source, which mechanically prevents direct estimation of the RDID interaction when source fixed effects are included in the regression. We address this constraint in the following section.

Standardization and the Z-Score Specification

The identification strategy developed in the preceding section establishes that source-level control is necessary to block the backdoor paths through country and ownership. However, including source fixed effects in the RDID regression is infeasible: because each source belongs to exactly one group (state-owned or independent, Russia-aligned or

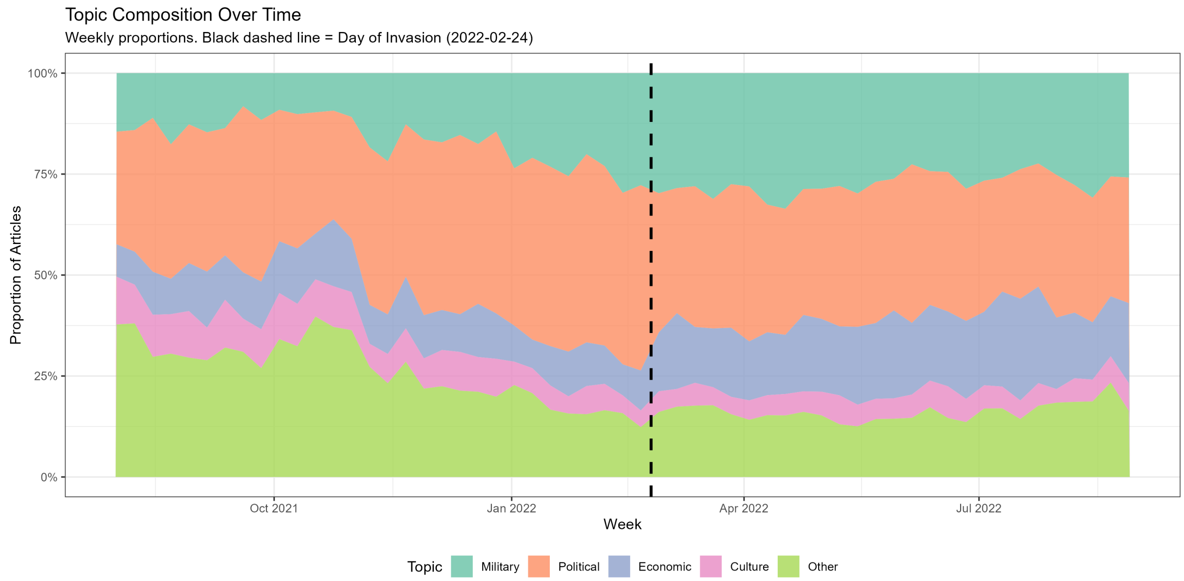

Figure 4: This figure shows the proportion of articles each topic comprises over time, measured weekly. Political topics dominate the pre-invasion period, but leading up to the invasion, military articles begin to grow in prominence. Other articles experience a steady decline before the invasion, but are flat afterwards. After the invasion, there is an increase in economic articles that draws from political articles.

neutral), source fixed effects perfectly absorb the group indicator G_p and mechanically prevent estimation of the RDID interaction β_5.

We resolve this constraint by standardizing the dependent variable at the source level prior to estimation. For each source s, we compute:

$$Y_p^z = \frac{Y_p - \bar{Y}_s}{\sigma_s} \tag{4}$$

where $\bar{Y}_s$ and σ_s are the mean and standard deviation of sentiment for source s, computed over the full sample period. The demeaning component $(Y_p - \bar{Y}_s)$ is mathematically equivalent to a source fixed effect by the Frisch-Waugh-Lovell theorem (Lovell, 1963): it removes all source-level intercept differences, including those attributable to country, ownership, and baseline editorial orientation. Critically, because the group indicator is absorbed by demeaning rather than by a fixed effect in the regression, G_p remains estimable in the RDID specification, and the interaction β_5 can be identified.

The division by σ_s addresses a second concern. Although all articles in our sample are coded on a ternary sentiment scale (-1, 0, 1), sources differ substantially in baseline variance—some produce predominantly neutral coverage while others are more polarized. In a linear specification, high-variance sources receive disproportionate influence on the estimated coefficients (Aronow & Samii, 2016). Without adjustment, the RDID estimate would be biased toward the experience of whichever sources happen to exhibit the greatest variation in sentiment, rather than reflecting the average response across sources. Dividing by σ_s ensures that each source contributes proportionally to the estimate, regardless of its baseline variance. It also aids interpretation: coefficients from the z-scored specification represent shifts in within-source standard deviations, providing a natural scale for comparing effect magnitudes across groups with different baseline distributions.

Two features of this standardization merit emphasis. First, the z-score is computed over the *full sample period*, not within the estimation bandwidth. This is a deliberate choice: if the standardization were computed within the bandwidth, the mean and variance would themselves be functions of the bandwidth, complicating interpretation and introducing a dependency between bandwidth selection and the dependent variable. Computing the z-score over the full sample period ensures that the standardization is a fixed transformation of the data, independent of the estimation procedure.

Second, we validate the z-score specification against the full fixed-effects approach in the one setting where both are estimable. For hypothesis 1—the within-group RDD—we first estimate the discontinuity using `rdrobust` with the complete set of fixed effects (country, source, and topic). We then re-estimate hypothesis 1 using the z-scored outcome with topic fixed effects only. The two approaches produce directionally equivalent results, confirming that the z-score specification adequately absorbs source-level heterogeneity. Having established this equivalence in the setting where both approaches are available, we proceed with the z-score specification for hypotheses 2 and 3, where the full fixed-effects structure cannot accommodate the group interaction.

The Forcing Variable

The forcing variable in our design is the distance between an article's publication date-time and the *date* of invasion (February 24^{th}, 2022). We measure distance from the date rather than a specific moment of invasion to maintain a sharp design: all articles published on or after February 24^{th} are treated, with the within-day publication time providing variation in proximity to the cutoff.[7] A standard approach in time-based RDD would use calendar date alone as the running variable. However, calendar date creates severe probability mass points: hundreds of articles may share the same publication date, producing substantial heaping that complicates both estimation and inference (Cattaneo et al., 2024). Incorporating publication time effectively eliminates mass points—exact timestamp ties are rare, rendering the forcing variable continuous or very nearly so.

Not all articles in our corpus have observed publication times. For these articles, we impute a publication time using kernel density estimation over the distribution of observed publication times for the same source. If a source has too few observed timestamps to estimate a reliable density, we instead use the country-level distribution of publication times. This approach preserves the publishing rhythm of each source as a prior: an article from a source that typically publishes in the morning will receive a morning timestamp, not a time drawn from a uniform distribution. Each article receives a single imputed time, drawn once and held fixed across all analyses.

We subject this approach to two robustness checks, reported in the appendix. First, we replicate all analyses using publication date rather than date-time. Mass points are present in this specification, but results are substantively identical to the date-time results, suggesting that heaping affects precision but does not introduce bias. Second, we replicate all analyses after dropping articles without observed publication times entirely. Results again hold, confirming that the imputation procedure is not driving our findings.

[7] Using a specific moment (e.g., the hour the invasion began) would require treating articles published on February 24^{th} before that moment as untreated, introducing fuzziness in treatment assignment.

Estimation and Inference

Our estimation procedures differ across the two designs. For hypothesis 1, we estimate the within-group RDD using `rdrobust` (Cattaneo et al., 2019), which implements local polynomial regression with MSE-optimal bandwidth selection (Calonico et al., 2020), bias-corrected confidence intervals, and cluster-robust standard errors at the source level.

For hypotheses 2 and 3, no nonparametric framework for the RDID currently exists, and the fully interacted specification with topic fixed effects exceeds what `rdrobust` can accommodate. We therefore estimate the RDID using OLS, which requires selecting a bandwidth separately. Because `rdbwselect` also cannot handle the fixed-effects structure directly, we first residualize the z-scored outcome on topic fixed effects and the group indicator using `feols`, then pass these residuals to `rdbwselect` with a triangular kernel, source-level clustering, a local linear polynomial ($p = 1$), and the MSE-optimal selection rule. This ensures the bandwidth is selected on the variation that the RDID specification actually exploits. Having obtained the optimal bandwidth, we filter the data to observations within this window and estimate the RDID via `feols` using the specification in Equation 3. Results are robust across bandwidths ranging from half to approximately 1.5 times the optimal bandwidth (see appendix).

Inference with Few Clusters

A central challenge in our design is that the number of clusters—sources—is small. For the full-sample RDID (hypothesis 2), we have approximately 60 source-level clusters. For the state-media-only RDID (hypothesis 3), we have 13. Standard cluster-robust variance estimators assume that the number of clusters is large; with few clusters, they can substantially over- or under-reject (Cameron & Miller, 2015).

We address this using the wild cluster restricted (WCR) bootstrap of Cameron et al. (2008) with Webb six-point weights and 4,999 replications, imposing the null hypothesis. These are preferred to the default Rademacher weights when the number of clusters is small (MacKinnon & Webb, 2017), as in our hypothesis 3 sample ($G = 13$).

The WCR bootstrap provides asymptotic refinement relative to cluster-robust standard errors when the number of clusters is small, and the restricted version (imposing the null) has better finite-sample properties than the unrestricted alternative (MacKinnon & Webb, 2017). We report the bootstrap p-values for all RDID specifications. For the `rdrobust` models (hypothesis 1), inference relies on the package's internal cluster-robust standard errors at the source level.

As a further check on inference, we estimate fully saturated source-by-topic RDDs in the appendix—a separate regression for each source-topic cell—and aggregate using both a pairs bootstrap at the source level and a Bayesian bootstrap with Dirichlet-distributed weights (Oganisian & Roy, 2021; Rubin, 1981). Because each cell constitutes its own regression, this approach sidesteps the clustering problem entirely, though at the cost of reduced power. Results are consistent with the main specifications.

Finally, we note a consideration raised by Abadie et al. (2022): if the sources in our sample constitute the population of relevant outlets in these countries rather than a sample drawn from a larger superpopulation, then clustering may be overly conservative. Our sample is close to this setting—we collect all major outlets, not a random draw from a larger universe of potential sources—but we do not rely on this argument and report clustered inference throughout.

Interpreting the RDID

The within-group RDD estimates for hypothesis 1 are causal under the standard continuity assumption: absent the invasion, sentiment would have evolved smoothly through the cutoff. We have discussed the grounds for this assumption in the Regression Discontinuity Design section. The interpretive question that requires further discussion is what the RDID estimates for hypotheses 2 and 3 identify.

The RDID compares the magnitude of the discontinuity in one group to the magnitude of the discontinuity in another. As Bansak and Nowacki (2022) emphasize in the context of RDD heterogeneity, it is important to distinguish between estimates that are *conditional on* group membership and estimates that are *causally driven by* group membership. The former—the empirical fact that Russia-aligned state media responded differently from neutral state media—is identified under the parallel-discontinuities assumption described in the RDID section. The latter—the claim that Russia-alignment itself caused the differential response—requires stronger assumptions.

Tramontin Shinoki et al. (2024) develop the formal conditions under which the RDID supports causal identification, building on the design introduced by Grembi et al. (2016). The key insight is that the RDID is causally identified when it can be expressed as a *discontinuity of differences* rather than a *difference of discontinuities*. The distinction hinges on data structure. When the same units are observed across two periods—one in which a pre-existing policy already produces a discontinuity at a threshold, and a second in which a new policy is introduced at the same threshold—a standard RDD in the second period alone would conflate the effects of both policies. But because the same units are observed in both periods, the researcher can difference out each unit's period 1 outcome, removing the pre-existing discontinuity. The RDD estimated on these within-unit differences then isolates the effect of the new policy. This is the structure in Grembi et al.'s application: Italian municipalities observed before and after the relaxation of fiscal rules, where the same population threshold that determines the new treatment also determined prior fiscal constraints. Toral (2024) employ the same logic in political science, using within-school variation across periods to identify how political turnover differentially affects appointed and unappointed bureaucrats at a performance threshold.

Our setting does not have this panel structure. We do not observe the same units at two different discontinuities; we observe different groups—Russia-aligned and neutral state media—at a single discontinuity. The comparison is therefore a difference of discontinuities: separate RDDs estimated within each group, then compared. Without within-unit differencing to absorb pre-existing group differences at the cutoff, the RDID cannot rule out that the differential response reflects a factor correlated with group membership rather than the mechanism our theory describes.

We therefore interpret our RDID estimates as conditional on group membership—

the differential response is a robust empirical pattern, not a causal effect of Russia-alignment per se. Three features of our analysis, however, provide evidence that the pattern is driven by the mechanism our theory describes rather than by an omitted confounder correlated with group membership. First, a sensitivity analysis using the framework of Cinelli and Hazlett (2020), applied after coarsened exact matching on publication date and topic, yields a robustness value of 1.9%; as we detail in the Robustness section, nullifying the differential would require a confounder more strongly associated with sentiment than the invasion itself. Second, our theoretical framework generates opposing predictions for the signaling and domestic audience management accounts, and the data are consistent with the signaling account and inconsistent with the domestic account. An omitted confounder would need to not only produce the observed differential but also do so in the specific direction that contradicts our theory. Third, we test an alternative group partition—splitting states by the size of their ethnic Russian population rather than by Russia-alignment—and find no differential response (Table 6). If the RDID result were driven by a confounder correlated with proximity to Russia or cultural ties rather than by the alliance constraint our theory identifies, we would expect the ethnic Russian partition to produce a similar pattern. It does not.

Analysis

We present results sequentially, beginning with the baseline effect of the invasion on media sentiment across the region (H1), then examining divergence between state-owned and independent media (H2), and finally testing whether the pattern of divergence varies with formal security alignment (H3).

H1: The Invasion as a Negative Shock to Media Sentiment

Table 1 presents our RDD estimates of the effect of the invasion on sentiment toward Russia across all media in the sample. Model 1, the simplest specification without covariates, estimates a decline in raw sentiment of 0.130 ($p < 0.001$). Model 2 adds topic fixed effects; Model 3 further adds source fixed effects; Model 4 includes topic, source, and state fixed effects. Across these specifications, estimates range from 0.106 to 0.115, all significant at $p < 0.001$. The stability of the estimate across these specifications indicates that the result is not driven by differences across sources, countries, or topic composition.

Models 5 through 7 use the z-scored outcome, which centers sentiment at the source level and scales by within-source standard deviation, providing a standardized measure of the shift. Model 5 estimates a decline of 0.217 within-source standard deviations ($p < 0.001$). Model 6 omits articles classified as military in topic, testing whether the result is an artifact of the mechanical influx of negatively-valenced war coverage after February 24$^{\text{th}}$. The estimate attenuates slightly to 0.184 standard deviations but remains highly significant, confirming that the sentiment shift extends beyond military reporting to political, economic, cultural, and other coverage. Model 7 drops NATO-allied outlets from the sample; the result is robust to their exclusion.

Figure 5 presents the discontinuity visually for the non-military sample, corresponding to the Model 6 specification. The break at February 24$^{\text{th}}$ is clearly visible, with mean daily sentiment dropping from approximately -0.15 pre-invasion to approximately -0.30 immediately after. The post-invasion trend slopes upward, consistent with a gradual recovery in sentiment over the following months, but the initial discontinuity is unambiguous. Additional robustness checks, including bandwidth sensitivity and alternative forcing variable constructions, are reported in Table A1.

Table 1: H1 RDD — Effect of Russian Invasion on Media Sentiment (Probabilistic Time)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Invasion	-0.130***	-0.115***	-0.106***	-0.106***	-0.217***	-0.184***	-0.284***
	(0.022)	(0.021)	(0.015)	(0.015)	(0.036)	(0.033)	(0.038)
Topic FEs	-	X	X	X	X	X	X
State FEs	-	-	-	X	-	-	-
Source FEs	-	-	X	X	-	-	-
Source Z-Score	-	-	-	-	X	X	X
Clusters	61	61	61	61	61	61	45

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. Models 1-4 use the optimal bandwidth. Models 5-7 use a Z-Score adjustment instead of fixed effects. Model 6 omits Military Topics. Model 7 omits NATO allies.

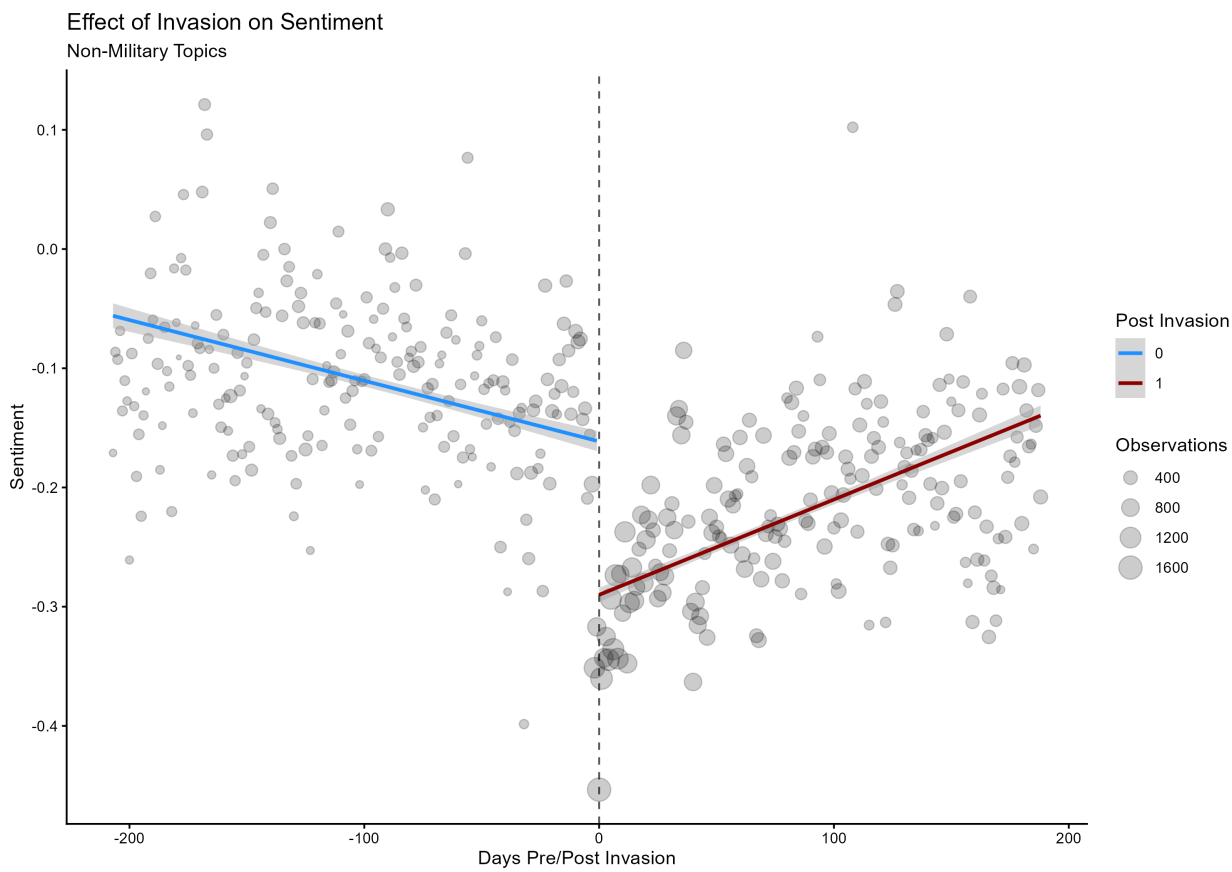

Figure 5: RDD estimate of the effect of the invasion on sentiment toward Russia, non-military articles. Each point represents one day, sized by the number of articles published. The dashed line marks February 24$^{\text{th}}$, 2022.

H2: State-Owned Media Manages Coverage More Carefully

Having established that the invasion produced a sharp negative shift in sentiment across the region, we now test whether state-owned and independent media responded differently. Hypothesis 2 predicts that state-owned media, operating under political constraints that independent outlets do not face, will produce a smaller decline in sentiment toward Russia.

Table 2 presents results in two stages. Models 1 and 2 report separate RDD estimates for state-owned and independent media, restricted to non-military articles. Independent media exhibits a decline of 0.214 within-source standard deviations ($p < 0.001$), while state-owned media declines by 0.109 standard deviations ($p < 0.1$). Both shift negatively, but independent media responds roughly twice as strongly as state-owned media.

Models 3 through 6 estimate the RDID interaction directly. The state-owned $\times$ treatment coefficient captures the extent to which state media's response to the invasion differs from that of independent media. Model 3, using the full sample at the optimal bandwidth, estimates an interaction of 0.080—positive, indicating that state media was less negative, but imprecise. Model 4 omits military articles, sharpening the estimate to 0.14 ($p < 0.1$). Models 5 and 6 restrict the sample to non-NATO countries, removing outlets whose institutional alignment with the West provides a distinct basis for opposing the invasion that is unrelated to the state-versus-independent dynamic our theory describes. In this sample, the interaction reaches 0.222 ($p < 0.05$), and the result is unchanged when coarsened exact matching on publication date and topic is applied (Model 6). The matched specification confirms that the divergence is not driven by observable differences in the timing or subject matter of coverage across state-owned and independent outlets.

Figure 6 presents the discontinuity by media type for the non-military sample. The visual pattern is consistent with the regression estimates: both state-owned and independent media shift downward at the cutoff, but independent media drops more sharply and recovers more slowly. The gap between the two post-invasion trend lines reflects the RDID interaction.

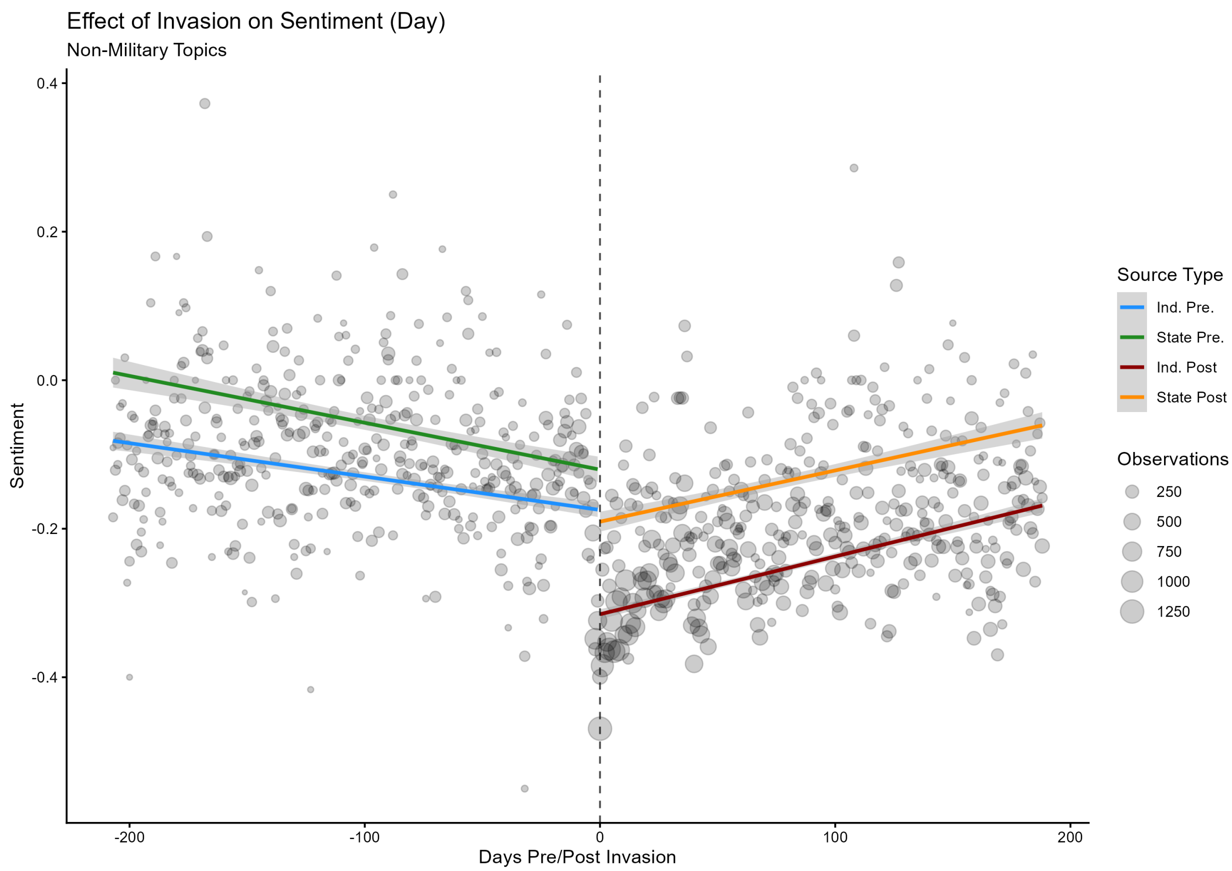

Figure 6: RDD estimates of the effect of the invasion on sentiment toward Russia by media type, non-military articles. Each point represents one day, sized by the number of articles published. The dashed line marks February 24th, 2022.

A single pair of articles illustrates the divergence Hypothesis 2 estimates and offers a spot-check on the classifier. In the days around the invasion, Serbia's state broadcaster and an independent outlet covered the same war from the same country, but their coverage of Russia pulled in opposite directions. On February 25, the independent outlet N1 ran "Canak: Putin started the attack on the same day that Hitler presented his program," reporting a Serbian politician's charge that Putin had launched the invasion on the anniversary of Hitler's Nazi program—coverage the classifier coded negative.[8] A day later, the state broadcaster RTS ran "Bocan-Harchenko: Russia understands

[8] Translations are machine-generated and reproduce the title and opening sentence supplied to the classifier; the source texts are in Serbian.

Table 2: H2 Diff-in-Disc — State Ownership Interaction, Z-Score (Probabilistic Time)

	(1) State-Owned‡	(2) Ind.‡	(3)†	(4)†	(5)†	(6)†
Invasion	-0.1090+	-0.214***	-0.253***	-0.240***	-0.347***	-0.348***
	(0.0583)	(0.034)	(0.036)	(0.031)	(0.027)	(0.026)
State-Owned x Treat			0.080	0.14+	0.222*	0.222*
			(0.063)	(0.06)	(0.069)	(0.069)
Source Z-Score	X	X	X	X	X	X
Clusters	17	44	60	60	43	43
Num.Obs.	30978	99510	40710	29278	31405	31379
R2			0.045	0.028	0.024	0.024
R2 Adj.			0.045	0.028	0.023	0.023

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All models use a source-level Z-Score normalization. Models 1-2 show separate RDD estimates for state-owned and independent media, filtered to omit military topics. Models 3-6 estimate the state-owned x treatment interaction. Model 3 uses the optimal bandwidth. Model 4 omits military topics. Model 5 omits NATO allies. Model 6 uses exact matching for topic and publication date on the Non-NATO data. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=4999, null imposed), clustered by source domain. ‡ Cluster-robust SEs from rdrobust.

Serbia's position, does not require anything from Belgrade," a report built around the Russian ambassador's assurances to Belgrade—coverage coded neutral. Both articles concern the same war, yet the classifier separates the one oriented negatively toward Russia from the one that is not. The medium is held constant—both are Serbian outlets covering the invasion in real time—so the gap in orientation tracks ownership rather than the event or the country's underlying disposition. This is the state-versus-independent divergence in miniature: the orientation of the coverage shifts with who controls the outlet, not with what there was to report.

Table 3 disaggregates the RDID by topic category. The state-owned × treatment interaction is positive across all five topics, indicating that the pattern of muted state media response is not confined to a single subject area. The interaction is statistically significant for economic coverage (0.29, $p < 0.05$) and other coverage (0.27, $p < 0.01$), and positive but imprecise for political (0.155), military (0.012), and cultural (0.10) topics. That the divergence is largest in economic coverage is suggestive, though we

Table 3: H2 Diff-in-Disc by Topic — Z-Score (Probabilistic Time)

	Political	Military	Economic	Cultural	Other
Invasion	-0.213***	-0.422***	-0.423***	-0.547***	-0.269***
	(0.027)	(0.068)	(0.066)	(0.071)	(0.039)
State Owned x Treatment	0.155	0.012	0.29*	0.10	0.27**
	(0.079)	(0.102)	(0.13)	(0.19)	(0.10)
Source Z-Score	X	X	X	X	X
Clusters	60	60	59	60	60
Num.Obs.	17390	18555	9491	3780	12671
R2	0.017	0.022	0.027	0.033	0.012
R2 Adj.	0.017	0.021	0.026	0.031	0.012

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
All models use a source-level Z-Score normalization with the optimal bandwidth. Each column restricts to the indicated topic category. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=4999, null imposed), clustered by source domain. ‡ Cluster-robust SEs from *rdrobust*.

do not theorize topic-level variation: states in our sample maintained substantial trade relationships with both Russia and the West, and economic framing may be the domain where strategic management of coverage is most consequential. The within-topic results also confirm that the H2 finding is not an artifact of compositional change—the divergence between state-owned and independent media is present within topic categories, not merely across them. Leave-one-out results and additional fixed-effects specifications are reported in Tables A3–A6.

H3: Russia-Aligned State Media Responds More Negatively than Neutral State Media

The central test of our argument compares state-owned media across security relationships. A domestic audience management account predicts that Russia-aligned states suppress negative coverage of their security patron, producing a smaller decline in sentiment than neutral states. Our signaling account predicts the opposite: because formal channels of dissent are foreclosed by the alliance constraint, Russia-aligned states use state media to signal distance from the patron to external audiences, producing a larger decline than neutral states whose formal channels remain open.

Table 4 presents the results. Models 1 and 2 report separate RDD estimates for Russia-aligned and neutral state media, restricted to non-military articles. Russia-aligned state media exhibits a significant decline of 0.220 within-source standard deviations ($p < 0.05$). Neutral state media, by contrast, shows no significant change (0.055, $p > 0.1$). The pattern is striking: it is the constrained states—those with formal security agreements with Russia, the same states that refused to condemn the invasion through diplomatic channels—whose state media shifted most negatively.

Models 3 through 5 estimate the RDID interaction directly. The Russia-aligned $\times$ treatment coefficient is negative across all specifications, indicating that Russia-aligned state media responded more negatively than neutral state media. Model 3, at the optimal bandwidth, estimates an interaction of -0.19, negative but imprecise. Model 4 omits military articles, sharpening the estimate to -0.33 ($p < 0.05$). Model 5 applies coarsened exact matching on publication date and topic to the non-military sample, producing an estimate of -0.38 ($p < 0.05$). The matched specification is our preferred estimate for this comparison: with only 13 source-level clusters, ensuring compositional equivalence across groups is particularly important, and matching on observables addresses the concern that the two groups differ in the timing or subject matter of their coverage. Inference for all RDID models uses the wild cluster restricted bootstrap with Webb six-point weights to account for the small number of clusters. A fully disaggregated analysis estimating separate RDDs within each source-topic cell, aggregated via both a pairs bootstrap and a Bayesian bootstrap, produces consistent results (Table A11).

Figure 7 presents the discontinuity by alignment group. Russia-aligned state media (blue pre-invasion, red post-invasion) exhibits a visible downward shift at the cutoff, while neutral state media (green pre-invasion, orange post-invasion) remains essentially flat. The divergence at the cutoff is the RDID estimate. The null result for neutral state media is not a floor effect; pre-invasion sentiment levels had substantial room to decline, as shown in Figure A10 in the appendix.

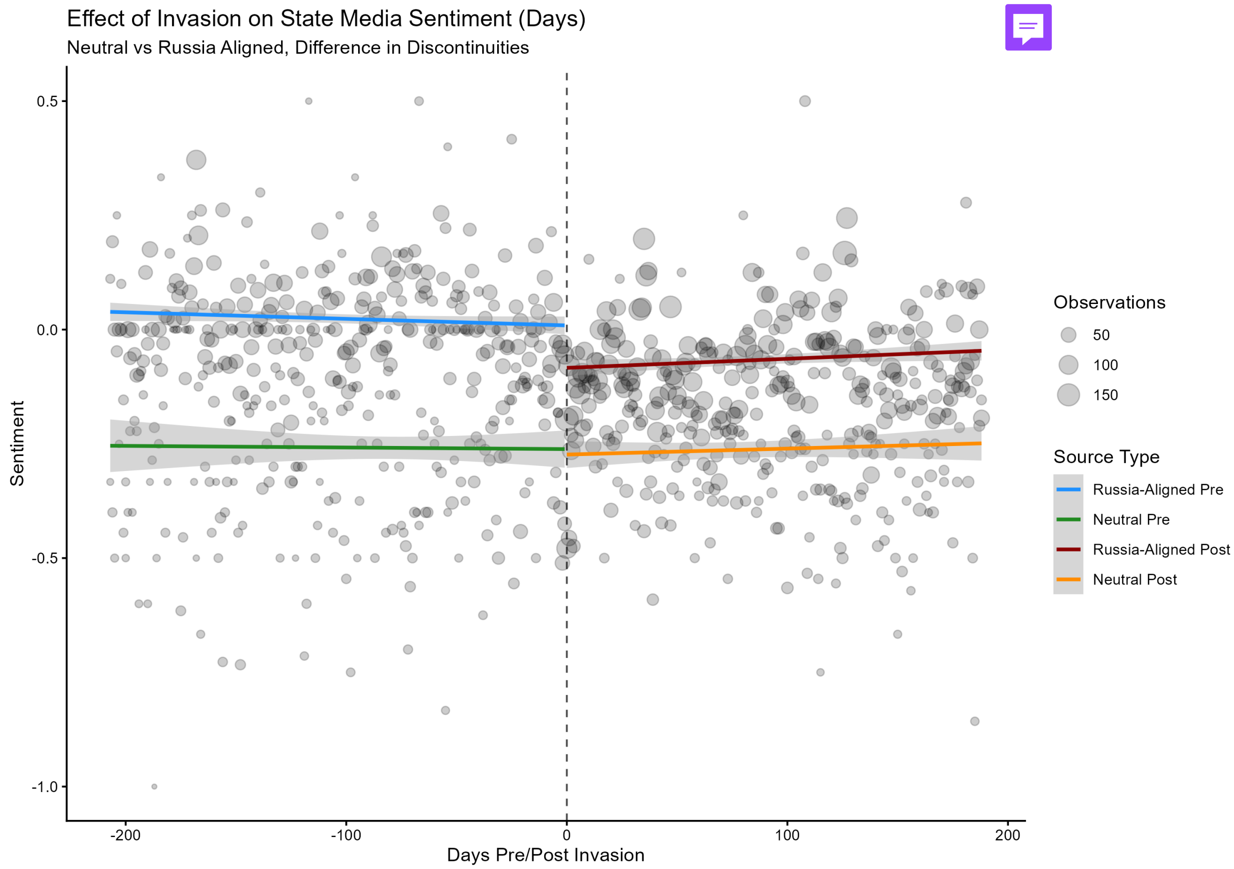

Figure 7: RDD estimates of the effect of the invasion on sentiment toward Russia in state-owned media, by security alignment. Sample restricted to state-owned media in non-NATO countries. Each point represents one day, sized by the number of articles published. The dashed line marks February 24$^{\text{th}}$, 2022.

The within-outlet shift in Armenia's state broadcaster illustrates the mechanism and shows it operating through selection rather than editorial voice. In January, the English-language service of Public Radio of Armenia carried "Opening of regional communications to give Armenia rail access to Russia, Europe, Iran, China," a report on the prime minister's infrastructure plans in which Russia figured as one economic partner among several—coverage the classifier coded positive toward Russia.[9] By March, the same outlet was running "Russia walks out of Council of Europe," foregrounding Russia's rupture with the European institutions of which Armenia is itself a member—coverage coded negative. Neither article is editorial commentary in the outlet's own

[9]Translations are machine-generated and reproduce the title and opening supplied to the classifier; the source is en.armradio.am.

Table 4: H3a Alignment Interaction — Z-Score (Probabilistic Time)

	(1) Russia-Aligned‡	(2) Neutral‡	(3)†	(4)†	(5)†
Invasion	-0.220*	0.055	-0.068	0.098	0.098
	(0.099)	(0.101)	(0.080)	(0.070)	(0.069)
Russia-Aligned x Treat			-0.19	-0.33*	-0.38*
			(0.11)	(0.11)	(0.11)
Source Z-Score	X	X	X	X	X
Clusters	9	4	13	13	13
Num.Obs.	17733	4802	7164	6094	5721
R2			0.025	0.015	0.014
R2 Adj.			0.024	0.013	0.013

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All models use a source-level Z-Score normalization. Sample restricted to state-owned media in non-NATO countries. Models 1-2 show separate RDD estimates for Russia-aligned and neutral countries, filtered to omit military topics. Models 3-5 estimate the Russia-Aligned x Treatment interaction. Model 3 uses the optimal bandwidth. Model 4 omits military topics. Model 5 uses exact matching on topic and date of publication for the non-military data. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=4999, null imposed), clustered by source domain. ‡ Cluster-robust SEs from *rdrobust*.

voice; the March piece is largely wire-style reporting of Russia's withdrawal and its combative justification. What changed is the orientation of the Russia coverage the outlet selected and foregrounded, from economic partnership to institutional rupture—and because it is a state outlet, that change in what it carries is readable as a deliberate choice rather than editorial drift. The shift occurred while Armenia withheld formal condemnation, abstaining on the UN General Assembly resolution against the invasion: the media-versus-diplomacy gap our account predicts for constrained states, captured in a single outlet.

The direction of this result provides leverage for distinguishing between the competing theoretical accounts. A domestic audience management account and our signaling account generate opposite predictions for hypothesis 3, and the data are consistent with the signaling account. Russia-aligned states did not suppress negative coverage to protect the alliance relationship; they permitted—or directed—a sharper negative shift in their state media than neutral states whose formal channels were unconstrained. As discussed in the Interpreting the RDID section, we interpret this differential as conditional on group membership rather than as a causal effect of Russia-alignment per se. Two features of the analysis—a sensitivity analysis bounding the role of unobserved factors and an alternative group partition—bear on whether the pattern reflects the mechanism our theory describes rather than an omitted confounder; we present both in the Robustness section below.

Robustness

We subject our results to a battery of robustness checks organized around the principal threats to identification. We present the sensitivity analysis and matching results in the main text; leave-one-out results, bandwidth variation, forcing variable alternatives, and the fully saturated model are reported in the appendix.

Sensitivity to Unobserved Confounding

The central threat to our RDID estimates is that an unobserved factor correlated with Russia-alignment produces a differential jump in sentiment at the cutoff through a channel other than the one our theory describes. We address this using a two-step procedure that combines coarsened exact matching with the sensitivity framework of Cinelli and Hazlett (2020).

We first construct a matched sample using coarsened exact matching (CEM; Iacus et al., 2012) on publication date and topic, pairing Russia-aligned and neutral articles that were published on the same day and assigned to the same topic category. Matching on these observable characteristics ensures that the RDID comparison is not driven by differences in the timing or subject matter of coverage across groups. We then estimate the RDID on the matched sample using the same specification as our main analysis (Equation 3), with the z-scored outcome, topic fixed effects, and the fully interacted running variable, treatment, and group indicator. The Cinelli and Hazlett (2020) framework, developed for OLS coefficients, applies directly to the RDID interaction coefficient ($T_p \times G_p$) in this specification.

We apply `sensemakr` to the matched-sample regression, designating the RDID interaction (`treatment:csto`) as the coefficient of interest, and use this confounding framework as a disciplined stress-test of the association rather than as a claim that the differential is a causal effect of group membership. The robustness value is 1.9%: an unobserved factor would need to explain at least 1.9% of the residual variance of both group assignment and sentiment to drive the estimated differential to zero.[10] To calibrate this, we benchmark against the invasion indicator itself—the strongest substantive predictor in the design. Even granted a near-mechanical 56% association with the Russia-aligned × treatment interaction, an unobserved factor would still need to be 1.5 times as strongly associated with sentiment as the invasion itself to drive the differential to zero.[11] Figure 8 presents the contour plot, with the dashed line marking the invasion indicator's own partial R^2 with sentiment to calibrate the vertical scale. Because our source fixed effects absorb all time-invariant outlet and country characteristics, the confounders that could satisfy this joint condition form a narrow class: they would have to vary differentially at the cutoff by group.

[10]We compute the robustness value using bootstrapped standard errors from the WCR bootstrap rather than the OLS standard errors from `lm()`, which do not account for clustering. The OLS-based robustness value would overstate the degree of robustness.

[11]The main treatment indicator is a component of the interaction $T_p \times G_p$, so at full strength the benchmark is degenerate: $k_d = 1$ implies a confounder partial R^2 with the interaction of at least one, which the procedure cannot evaluate. We therefore set $k_d = 0.25$; even at a quarter of the benchmark's strength the implied confounder must explain 56% of the interaction's residual variance. The partial R^2 with sentiment required to drive the differential to zero at that point is 0.0015, against the invasion indicator's own 0.0010.

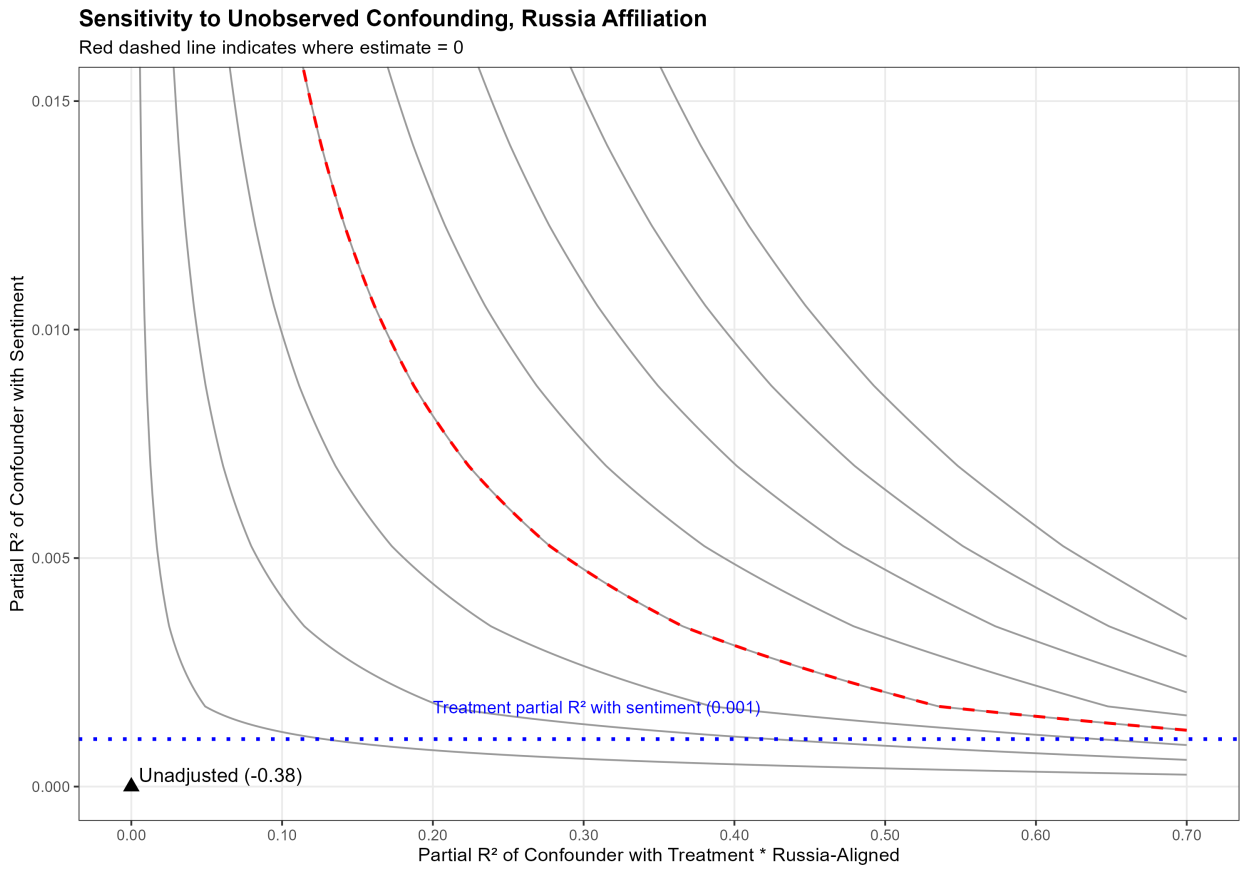

Figure 8: Sensitivity of the Russia-aligned × treatment interaction to unobserved confounding, following Cinelli and Hazlett (2020). The red dashed line indicates the combinations of confounder strength that would drive the estimated differential to zero. The triangle marks the unadjusted estimate; the dashed horizontal line marks the invasion indicator's own partial R^2 with sentiment (≈ 0.001), calibrating the vertical scale.

To our knowledge, cross-group RDID comparisons of this kind are not routinely subjected to formal sensitivity analysis; we propose this two-step procedure—coarsened exact matching to address observable compositional differences across groups, paired with the Cinelli and Hazlett (2020) framework to bound the role of unobservables—as a standard robustness protocol for the setting. Establishing such tests for associational comparisons is itself a contribution: as Lieberman (2016) argues, the descriptive and correlational stages of inquiry merit the same methodological care as causal ones, and a differential shown robust to this scrutiny is a firmer foundation for the causal work that might follow.

Table 5: Ethnic Russian Population x Russia Security Alignment
Country Classification

	Low Alignment (Neutral)	High Alignment (Rus. Align.)
Low Ethnic	Kosovo, Serbia	Azerbaijan, Armenia
High Ethnic	Moldova, Georgia	Belarus, Kazakhstan

An Alternative Group Partition

We test an alternative group partition that provides a direct check on a prominent competing explanation. If the differential response were driven by cultural proximity to Russia rather than by the alliance constraint, we would expect a similar pattern when states are partitioned by the share of ethnic Russian population—the dimension Russia itself invoked to justify the invasion. Table 5 presents the cross-classification of countries by Russia-alignment and ethnic Russian population share. The two dimensions cross-cut cleanly: Azerbaijan and Armenia are Russia-aligned but have low ethnic Russian populations, while Moldova and Georgia are neutral but have high ethnic Russian populations. This cross-cutting structure gives the test discriminating power: a result driven by cultural proximity would appear in the ethnic Russian partition regardless of Russia-alignment. The design's value lies in this orthogonality: alignment and ethnic-Russian share cut across our cases. Because the two dimensions cross-cut, they make divergent predictions the data can adjudicate. Because the partition is orthogonal to alignment, a result that loads on alignment but not on ethnic-Russian share isolates our proposed mechanism rather than a correlate of it.

Table 6 presents the results. Models 1 and 2 report separate RDD estimates for high and low ethnic Russian population countries; neither shows a significant discontinuity. Models 3 through 5 estimate the interaction directly. The high Russian population $\times$ treatment coefficient is substantively small and statistically insignificant across all specifications, ranging from -0.055 to 0.015. The null result is stark when compared to the Russia-aligned partition: the same sample, the same methodology, and the same estimation procedure produce a significant differential when countries are split by security alignment and no differential when split by ethnic composition. This pattern

Table 6: H3b Ethnic Russian Population — Null Result, Z-Score (Probabilistic Time)

	(1) High‡	(2) Low‡	(3)†	(4)†	(5)†
Invasion	-0.079	-0.108	-0.17	-0.13	-0.143
	(0.066)	(0.102)	(0.10)	(0.10)	(0.099)
High Russian Pop x Treatment			-0.055	-0.0024	0.015
			(0.133)	(0.1479)	(0.145)
Source Z-Score	X	X	X	X	X
Clusters	9	4	13	13	13
Num.Obs.	13942	8593	7202	6092	5992
R2			0.022	0.012	0.011
R2 Adj.			0.021	0.010	0.010

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
All models use a source-level Z-Score normalization. Models 1-2 show separate RDD estimates for high and low ethnic Russian population countries. Models 3-5 estimate the high Russian population x treatment interaction. Model 3 uses the optimal bandwidth. Model 4 omits military topics. Model 5 uses matched data. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=4999, null imposed), clustered by source domain. ‡ Cluster-robust SEs from *rdrobust*.

is inconsistent with the cultural proximity explanation and supports the interpretation that the mechanism driving the H3 result operates through the alliance constraint rather than through ethnic or cultural ties to Russia. Leave-one-out results and additional specifications are reported in Tables A7–A10.

Sample Composition

We assess whether our results are driven by any single country or outlet through leave-one-out analysis. We re-estimate all specifications after sequentially dropping each country and each source from the sample. Results are stable across all exclusions, reported in the appendix.

We also present matched results in the main text for both the state-owned versus independent comparison (hypothesis 2; Table 2) and the Russia-aligned versus neutral comparison (hypothesis 3; Table 4). As described above, matching is performed via CEM on publication date and topic. The matched estimates are consistent with the unmatched specifications, indicating that observable differences in article timing and subject matter across groups do not account for the differential response.

For hypothesis 2, we additionally restrict the sample to non-NATO countries. Because NATO members have a clear institutional basis for opposing the invasion that is distinct from the signaling mechanism we theorize, including them risks conflating alliance-driven opposition with the state-versus-independent media dynamic. Dropping NATO-affiliated outlets does not attenuate the result (Table 2).

Specification and Compositional Threats

As discussed in the Identification section, topic fixed effects may inadequately address the compositional threat if topic composition shifts discontinuously at the cutoff. We address this in two ways. First, we omit articles coded as military in nature and re-estimate all specifications. If the main result were driven by the mechanical influx of negatively-valenced military coverage, dropping these articles would attenuate or eliminate the effect. It does not: results hold across all hypotheses (Tables 2, 4). Second, we estimate the RDID within each topic category separately (Table 3), confirming that the differential response is present within non-military topics and is not an artifact of compositional change.

Placebo Test

We test the continuity assumption by estimating the RDD at a placebo cutoff of December 7$^{\text{th}}$, 2021—the date the United States publicly announced that Russia intended to invade Ukraine. If media sentiment were already shifting discontinuously in the weeks before the invasion, we would expect to detect a significant break at this date. We find no significant discontinuity, consistent with the interpretation that the invasion itself, rather than its anticipation, produced the sharp shift in coverage.

Additional specification checks—including alternative forcing variable constructions (publication date rather than date-time; dropping articles with imputed timestamps) and bandwidth variation (results stable from 0.5 to 1.5 times the optimal bandwidth)—are reported in the appendix.

Conclusion

The Russian invasion of Ukraine confronted Russia's security partners with a dilemma. These states had genuine reasons to oppose the invasion, but the directly attributable channels through which the discipline has primarily studied dissent—UN votes, official statements, public breaks—risked provoking the very patron that guarantees their security. Our findings show that these states turned to state-owned media. The invasion produced a sharp negative shift in sentiment toward Russia across the region (H1), and state-owned media managed this shift more carefully than independent media, consistent with strategic control of coverage (H2). That independent media responded more strongly than state media is itself evidence that state outlets operate under political constraints that shape their coverage—if both responded identically, there would be no strategic management to detect. The central finding, however, is that state-owned media in Russia-aligned states shifted more negatively than state-owned media in neutral states (H3)—the opposite of what a domestic audience management account predicts, and precisely the pattern our signaling argument describes. States whose formal channels were most constrained produced the sharpest media signal, despite these same states refusing to condemn the invasion through formal diplomatic channels. An alternative partition by ethnic Russian population share produced no differential, ruling out cultural proximity as a driver and providing further evidence that the mechanism operates through the alliance constraint.

These findings speak to a long-standing disconnect between international relations scholarship and intelligence practice. Western intelligence services have monitored state media as a window into regime preferences since the Cold War, treating shifts in tone and emphasis as indicators of factional dynamics and policy direction within regimes whose formal diplomatic output is tightly controlled. The discipline, by contrast, has studied state media almost exclusively as a tool of domestic audience management. Our analysis demonstrates that the signals practitioners monitor are not idiosyncratic to particular crises or analysts—they are theoretically structured and empirically identifiable. The patterns we document follow directly from the logic of alliance constraints, and they discriminate between competing theoretical accounts: the domestic management and signaling accounts generate opposite predictions for how Russia-aligned state media should respond to the invasion, and the data are consistent with the signaling account alone. State media sentiment, in short, is not noise. It is a legible instrument of international communication, one that the discipline can study with the same rigor it applies to formal diplomatic behavior.

Our second contribution identifies the channel through which this communication operates. The signaling literature has focused on how states communicate resolve or commitment through directly attributable instruments—UN votes, official statements, military deployments—precisely because their visibility generates the costs that make them credible. We show that when these channels are foreclosed by alliance constraints, states turn to institutionally mediated ones. State media has just this character: tied closely enough to the government that a shift in its coverage reads as intentional, but not so formal that the shift becomes an act of state. The signal is costly enough to shift the receiver's beliefs about the sender's alignment, but institutionally buffered—through the structural separation between government and editorial function—to avoid crossing the threshold that has historically triggered patron retaliation. This logic is not unique to the present case. Any setting in which a subordinate actor depends on a patron but must manage relationships with other audiences creates pressure for signaling through channels that are observable to the intended audience but do not commit the sender to an official position. It should extend beyond international relations to any patron-client hierarchy—subnational governments constrained by central authorities, or corporate subsidiaries navigating the preferences of a parent firm—though we leave these extensions to future work. The same pressure is strongest where patron and client are mutually dependent—whether through security commitments, economic ties, or both—such that the patron's cost of punishing media-level dissent exceeds the benefit of enforcing message discipline.

Within international relations, natural extensions include states in China's security orbit today, US-backed regimes in Central America during the 1980s and 1990s, where similar dynamics of patron dependence and constrained dissent shaped the relationship between state media and official policy, and contemporary cases in which US allies navigate the tension between security dependence and domestic or international pressure regarding conflicts such as the war in Gaza. The 2022 invasion provides unusual analytical clarity—a discontinuity, a well-defined alliance structure, and competing theoretical predictions—but the underlying mechanism should operate wherever directly attributable channels of dissent are foreclosed by dependence on a powerful patron.

Methodologically, our approach offers two contributions. First, the pairing of prompted large language model classification with a regression discontinuity design provides a template for large-scale analysis of media signals in contexts where hand-coding is infeasible. The prompted LLM approach is not incidental to the measurement strategy: by specifying sentiment toward Russia as the classification target, we measure the article's orientation toward a specific actor rather than the overall tone of the coverage, a distinction that off-the-shelf sentiment classifiers cannot make. Second, our combination of coarsened exact matching with the sensitivity framework of Cinelli and Hazlett (2020) offers a transparent, reusable procedure for assessing how robust a cross-group comparison is to omitted factors when group membership is observational rather than assigned—a common setting in comparative research that is rarely subjected to formal sensitivity analysis. Disciplining associational comparisons in this way is, following Lieberman (2016), valuable in its own right: it strengthens the descriptive and correlational foundations on which later causal work depends.

Three limitations merit discussion. First, our findings are drawn from a single crisis. The invasion of Ukraine provides unusual analytical clarity, but the generalizability of the results depends on whether the same logic operates in other settings where directly attributable channels are constrained by patron dependence. Testing this across other alliance structures is a priority for future work. Second, we document sender behavior but not receiver response. The institutional infrastructure for monitoring state media is well-established—Western intelligence services have done so systematically for decades—and states in our sample operate in an environment where they can reasonably expect their coverage to be observed. But we do not trace the signal from sender to receiver, and we cannot confirm that the shifts we identify were detected, interpreted, or acted upon by the intended audience. Third, as we discuss in the theory section, our data cannot distinguish whether the state media shifts reflect top-down strategic direction by regime leadership or the expression of elite factional pressure through outlets that different factions influence. Both mechanisms produce the same observable pattern, and the divergence itself—that the regime is not unified in its public posture toward its security patron—is analytically meaningful regardless of which internal process produces it.

The sender-receiver gap points toward productive avenues for future research. Two approaches could help close the loop. First, examining whether Western sanctions targeted Russia-aligned states differentially based on their media posture would provide evidence on whether the signals we identify correlate with policy responses. Second, analyzing whether publicly traded firms in Russia-aligned countries with deep Western ties experienced financial penalties correlated with state media sentiment shifts would test whether markets—another audience that monitors public information—responded to the same signals. Neither measure perfectly captures signal receipt, but both would move the analysis from documenting the sender's behavior to tracing its consequences. More broadly, the theoretical framework developed here provides a basis for interpreting what intelligence practitioners have long observed. Western governments have monitored state media as a source of regime preferences for decades; this paper offers a theoretical account of why the practice works. The signals are not ad hoc—they follow from the structure of alliance constraints, which means they can be interpreted more systematically and, as we have demonstrated, quantified at scale. As the international system moves toward a more multipolar order, patron-client alliances will continue to press the communication bind upon smaller states. Understanding how they communicate under these conditions will be a fundamental component of studying international relations in the decades ahead.

References

Abadie, A., Athey, S., Imbens, G. W., & Wooldridge, J. M. (2022). When Should You Adjust Standard Errors for Clustering? *The Quarterly Journal of Economics*, *138*(1), 1–35. https://doi.org/10.1093/qje/qjac038

Allison, R. (2008). Russia resurgent? Moscow's campaign to 'coerce Georgia to peace'. *International Affairs*, *84*(6), 1145–1171. https://doi.org/10.1111/j.1468-2346.2008.00762.x

Aronow, P. M., & Samii, C. (2016). Does Regression Produce Representative Estimates of Causal Effects? *American Journal of Political Science*, *60*(1), 250–267. https://doi.org/10.1111/ajps.12185

Banco, E., Graff, G., Seligman, L., Toosi, N., & Ward, A. (2023). 'Something Was Badly Wrong': When Washington Realized Russia Was Actually Invading Ukraine. *Politico*. Retrieved June 1, 2026, from https://www.politico.com/news/magazine/2023/02/24/russia-ukraine-war-oral-history-00083757?nlid=630318

Bank, W. (2021). World Integrated Trade Solution (WITS). https://wits.worldbank.org/#

Bansak, K., & Nowacki, T. (2022, June). Effect Heterogeneity and the Challenges of Causal Attribution in Regression Discontinuity Designs. https://doi.org/10.31235/osf.io/vj34m

BBC Monitoring. (2016, September). *Written Evidence to the House of Commons Defence Committee (BBC0002)* (Government Report). House of Commons. Retrieved March 18, 2026, from https://committees.parliament.uk/writtenevidence/72975/html/

Blankenship, B., & Lin-Greenberg, E. (2022). Trivial Tripwires?: Military Capabilities and Alliance Reassurance. *Security Studies*, *31*(1), 92–117. https://doi.org/10.1080/09636412.2022.2038662

Bueno de Mesquita, B. (Ed.). (2003). *The logic of political survival*. MIT Press.

Calonico, S., Cattaneo, M. D., & Farrell, M. H. (2020). Optimal bandwidth choice for robust bias-corrected inference in regression discontinuity designs. *The Econometrics Journal*, *23*(2), 192–210. https://doi.org/10.1093/ectj/utz022

Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). Bootstrap-Based Improvements for Inference with Clustered Errors. *The Review of Economics and Statistics*, *90*(3), 414–427. Retrieved March 20, 2024, from https://ideas.repec.org//a/tpr/restat/v90y2008i3p414-427.html

Cameron, A. C., & Miller, D. L. (2015). A Practitioner's Guide to Cluster-Robust Inference. *Journal of Human Resources*, *50*(2), 317–372. https://doi.org/10.3368/jhr.50.2.317

Cattaneo, M. D., Idrobo, N., & Titiunik, R. (2024, April). *A Practical Introduction to Regression Discontinuity Designs: Extensions* (1st ed.). Cambridge University Press. https://doi.org/10.1017/9781009441896

Cattaneo, M. D., Idrobo, N., & Titiunik, R. (2019, November). *A Practical Introduction to Regression Discontinuity Designs: Foundations* (1st ed.). Cambridge University Press. https://doi.org/10.1017/9781108684606

Cattaneo, M. D., Keele, L., & Titiunik, R. (2023, March). Covariate Adjustment in Regression Discontinuity Designs. In *Handbook of Matching and Weighting Adjustments for Causal Inference* (1st ed., pp. 153–168). Chapman; Hall/CRC. https://doi.org/10.1201/9781003102670-8

Cattaneo, M. D., & Titiunik, R. (2022). Regression Discontinuity Designs. *Annual Review of Economics*, *14*(1), 821–851. https://doi.org/10.1146/annurev-economics-051520-021409

Cheterian, V. (2009). The August 2008 war in Georgia: From ethnic conflict to border wars. *Central Asian Survey*, *28*(2), 155–170. https://doi.org/10.1080/02634930903056768

Chong, D., & Druckman, J. N. (2007). Framing Public Opinion in Competitive Democracies. *American Political Science Review*, *101*(4), 637–655. https://doi.org/10.1017/S0003055407070554

Cinelli, C., & Hazlett, C. (2020). Making Sense of Sensitivity: Extending Omitted Variable Bias. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *82*(1), 39–67. https://doi.org/10.1111/rssb.12348

Cohen, R. (1987). *Theatre of power: The art of diplomatic signalling*. Longman.

Dylan, H., & Maguire, T. J. (2022). Secret Intelligence and Public Diplomacy in the Ukraine War. *Survival*, *64*(4), 33–74. https://doi.org/10.1080/00396338.2022.2103257

Egorov, G., Guriev, S., & Sonin, K. (2009). Why Resource-poor Dictators Allow Freer Media: A Theory and Evidence from Panel Data. *American Political Science Review*, *103*(4), 645–668. https://doi.org/10.1017/S0003055409990219

Enikolopov, R., Petrova, M., & Zhuravskaya, E. (2011). Media and Political Persuasion: Evidence from Russia. *American Economic Review*, *101*(7), 3253–3285. https://doi.org/10.1257/aer.101.7.3253

Entman, R. M. (2004). *Projections of power: Framing news, public opinion, and U.S. foreign policy* (Nachdr.). University of Chicago Press.

Favere, A. (2024). Debating Open Source: A Practitioner's Perspective. *Studies in Intelligence*, *68*(3). Retrieved March 18, 2026, from https://www.cia.gov/resources/csi/studies-in-intelligence/studies-in-intelligence-68-no-3-extracts-september-2024/debating-open-source-a-practitioners-perspective/

Fearon, J. D. (1997). Signaling Foreign Policy Interests: Tying Hands versus Sinking Costs. *Journal of Conflict Resolution*, *41*(1), 68–90. https://doi.org/10.1177/0022002797041001004

Flamer, N. (2023). 'The enemy teaches us how to operate': Palestinian Hamas use of open source intelligence (OSINT) in its intelligence warfare against Israel (1987-2012). *Intelligence and National Security*, *38*(7), 1171–1188. https://doi.org/10.1080/02684527.2023.2212556

Fuhrmann, M., & Sechser, T. S. (2014). Signaling Alliance Commitments: Hand-Tying and Sunk Costs in Extended Nuclear Deterrence. *American Journal of Political Science*, *58*(4), 919–935. https://doi.org/10.1111/ajps.12082

Gandhi, J. (2008). *Political institutions under dictatorship*. Cambridge University Press.

Gehlbach, S., & Sonin, K. (2014). Government control of the media. *Journal of Public Economics*, *118*, 163–171. https://doi.org/10.1016/j.jpubeco.2014.06.004

Gershkoff, A., & Kushner, S. (2005). Shaping Public Opinion: The 9/11-Iraq Connection in the Bush Administration's Rhetoric. *Perspectives on Politics*, *3*(3), 525–537. https://doi.org/10.1017/S1537592705050334

Götz, E., & Staun, J. (2022). Why Russia attacked Ukraine: Strategic culture and radicalized narratives. *Contemporary Security Policy*, *43*(3), 482–497. https://doi.org/10.1080/13523260.2022.2082633

Grembi, V., Nannicini, T., & Troiano, U. (2016). Do Fiscal Rules Matter? *American Economic Journal: Applied Economics*, *8*(3), 1–30. https://doi.org/10.1257/app.20150076

Guriev, S. M., & Treisman, D. (2022). *Spin dictators: The changing face of tyranny in the 21st century*. Princeton university press.

Hirschman, A. O. (1970). *Exit, voice and loyalty: Responses to decline in firms, organisations and States*. Harvard university press.

Huseynov, V. (2022). Azerbaijan and Russia Sign Declaration on Allied Cooperation. *Foreign Policy*, *19*(25). https://jamestown.org/azerbaijan-and-russia-sign-declaration-on-allied-cooperation/

Iacus, S. M., King, G., & Porro, G. (2012). Causal Inference without Balance Checking: Coarsened Exact Matching. *Political Analysis*, *20*(1), 1–24. https://doi.org/10.1093/pan/mpr013

Iyengar, S., & Kinder, D. R. (2010). *News that matters: Television and American opinion* (Updated ed). University of Chicago Press.

Jazeera, A. (2022, January). US and NATO deliver responses to Russian demands over Ukraine. Retrieved April 28, 2024, from https://www.aljazeera.com/news/2022/1/26/us-nato-deliver-responses-to-russian-demands-over-ukraine

Kassing, J. W. (1997). Articulating, antagonizing, and displacing: A model of employee dissent. *Communication Studies*, *48*(4), 311–332. https://doi.org/10.1080/10510979709368510

Lieberman, E. S. (2016). Can the Biomedical Research Cycle be a Model for Political Science? *Perspectives on Politics*, *14*(4), 1054–1066. https://doi.org/10.1017/S153759271600298X

Lovell, M. C. (1963). Seasonal Adjustment of Economic Time Series and Multiple Regression Analysis. *Journal of the American Statistical Association*, *58*(304), 993–1010. https://doi.org/10.1080/01621459.1963.10480682

MacKinnon, J. G., & Webb, M. D. (2017). Wild Bootstrap Inference for Wildly Different Cluster Sizes. *Journal of Applied Econometrics*, *32*(2), 233–254. https://doi.org/10.1002/jae.2508

Malesky, E., & Schuler, P. (2010). Nodding or Needling: Analyzing Delegate Responsiveness in an Authoritarian Parliament. *American Political Science Review*, *104*(3), 482–502. https://doi.org/10.1017/S0003055410000250

Mankoff, J. (2022). Turkey's Balancing Act on Ukraine Is Becoming More Precarious. *Foreign Policy*. https://foreignpolicy.com/2022/03/10/turkey-ukraine-russia-war-nato-erdogan/

Manor, I. (2019). *The Digitalization of Public Diplomacy*. Palgrave Macmillan US.

McCombs, M. (2002). The agenda-setting role of the mass media in the shaping of public opinion. http://sticerd.lse.ac.uk/dps/extra/McCombs

Moratz, D., Springman, J., Wibbels, E., Adiguzel, F. S. S., Villamizar Chaparro, S. M., Lin, Z.-R., Romero, D., Soltani, M., Su, H., & Swami, J. (2025, August). Tracking Civic Space in Developing Countries with a High-Quality Corpus of Domestic Media and Transformer Models. https://doi.org/10.31235/osf.io/zp3sr_v2

Morrow, J. D. (1991). Alliances and Asymmetry: An Alternative to the Capability Aggregation Model of Alliances. *American Journal of Political Science*, *35*(4), 904. https://doi.org/10.2307/2111499

Nations, U. (1945, June). UN Charter. Retrieved April 23, 2024, from https://www.un.org/en/about-us/un-charter

Newman, E., & Visoka, G. (2016). The Foreign Policy of State Recognition: Kosovo's Diplomatic Strategy to Join International Society: Table 1. *Foreign Policy Analysis*, orw042. https://doi.org/10.1093/fpa/orw042

Oates, S. (2006). *Television, democracy and elections in Russia* [OCLC: 1345642721]. Taylor & Francis.

Oganisian, A., & Roy, J. A. (2021). A practical introduction to Bayesian estimation of causal effects: Parametric and nonparametric approaches. *Statistics in Medicine*, *40*(2), 518–551. https://doi.org/10.1002/sim.8761

Peisakhin, L., & Rozenas, A. (2018). Electoral Effects of Biased Media: Russian Television in Ukraine. *American Journal of Political Science*, *62*(3), 535–550. https://doi.org/10.1111/ajps.12355

Potter, B. (2022). Unrecognized Republic, Recognizable Consequences: Russian Troops in “Frozen” Transnistria. *Journal of Advanced Military Studies*, *2022*(special), 168–188. https://doi.org/10.21140/mcuj.2022SIstratcul010

Radeljić, B., & Özşahin, M. C. (2023). The inefficiency of EU leverage in Serbia during the Russia-Ukraine war. *Southeast European and Black Sea Studies*, *23*(4), 697–716. https://doi.org/10.1080/14683857.2023.2255029

Robinson, P. (2002). *The CNN Effect: The Myth of News, Foreign Policy and Intervention*. Taylor; Francis.

Rubin, D. B. (1981). The Bayesian Bootstrap. *The Annals of Statistics*, *9*(1). https://doi.org/10.1214/aos/1176345338

Slothuus, R., & Bisgaard, M. (2021). How Political Parties Shape Public Opinion in the Real World. *American Journal of Political Science*, *65*(4), 896–911. https://doi.org/10.1111/ajps.12550

Snyder, G. H. (1997). *Alliance politics*. Cornell University Press.

Svolik, M. W. (2012). *The politics of authoritarian rule*. Cambridge University Press.

Toral, G. (2024). How Patronage Delivers: Political Appointments, Bureaucratic Accountability, and Service Delivery in Brazil. *American Journal of Political Science*, *68*(2), 797–815. https://doi.org/10.1111/ajps.12758

Tramontin Shinoki, S., Picchetti, P., & Campos De Xavier Pinto, C. (2024). Difference-in-Discontinuities: Estimation, Inference and Validity Tests. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4846141

USIC. (2024, March). The IC OSINT Strategy, 2024-2026. https://www.dni.gov/files/ODNI/documents/IC_OSINT_Strategy.pdf

Walker, N. (2023, August). *Conflict in Ukraine: A timeline (2014 - eve of 2022 invasion)* (tech. rep. No. Number CBP 9476). House of Commons Library. https://researchbriefings.files.parliament.uk/documents/CBP-9476/CBP-9476.pdf

Walker, N. (2024, March). *Conflict in Ukraine: A timeline (current conflict, 2022 - present)* (tech. rep.). https://commonslibrary.parliament.uk/research-briefings/cbp-9847/

Weitsman, P. A. (2004). *Dangerous alliances: Proponents of peace, weapons of war*. Stanford University Press.

Williams, H., & Blum, I. (2018). *Defining Second Generation Open Source Intelligence (OSINT) for the Defense Enterprise*. RAND Corporation. https://doi.org/10.7249/RR1964

Appendix

Article Sentiment Examples

The following articles illustrate the three sentiment classifications used in our analysis. Each example shows the translated title and first two sentences, which served as the classifier input. Articles were selected to represent clear cases across different countries and media ownership types.

Positive Sentiment Toward Russia

Serbia	*danas.rs*	Independent	2022-02-07

Petković: Russia's help on the issue of Kosovo is invaluable.

"The director of the Government Office for Kosovo and Metohija, Petar Petković, assessed today that for Belgrade, Russia's support and assistance on the issue of Kosovo is invaluable. Petković said in a conversation with Russian Ambassador to Belgrade Aleksandar Bocan Kharchenko that pressure and repression against the Serbian people [continues]."

Belarus	*eng.belta.by*	State-owned	2022-04-12

Lukashenko: Belarus has something to contribute to Russia's might.

"Belarus and Russia will surely endure in the current circumstances. Belarus President Aleksandr Lukashenko expressed confidence about it during negotiations with President of Russia Vladimir Putin at the Vostochny Cosmodrome in Russia's Amur Oblast on 12 April."

Neutral Sentiment Toward Russia

Turkey	*sozcu.com.tr*	Independent	2022-04-19

Russia and Ukraine traded prisoners.

"Russia and Ukraine made a prisoner exchange. While Russia's 56th day of occupation was concentrated in the east of the country, the two countries reportedly made a prisoner exchange."

Armenia	*golosarmenii.am*	Independent	2022-04-01

Armenia abstained in the UNHRC vote on unilateral sanctions.

"Twenty-seven of the 47 countries, including Russia, China and India, voted in favour of the relevant resolution adopted at the meeting on Thursday, while 14 States, including the United States, the European Union, the United Kingdom and Ukraine, objected. Six delegations abstained in the vote, including Armenia."

Negative Sentiment Toward Russia

Turkey	*sozcu.com.tr*	Independent	2022-03-30

The Russian-Ukraine war... there are bodies in the streets of Irpin.

"The Russian-Ukraina war... there are bodies on the streets in Irpin, near Kiev. A promising statement from Russia following the talks in Istanbul yesterday, but the images taken at Irpin, which were at the bottom of Kiev, terrified the international public."

Serbia	*danas.rs*	Independent	2022-05-26

De-Ukrainization in Mariupol: Russians installed mobile TV propaganda, children abolished holidays.

"The Russian forces that occupied Mariupol have decided to cancel the summer vacation so that the students can master the Russian program, said today the mayor's adviser Petro Andryushchenko. Children will have classes in Russian language and literature, Russian history and mathematics all summer."

Classifier Performance

The main text reports validation results for the sentiment and topic classifiers. The Russian involvement classifier, which identifies whether Russia is a main actor in an article, achieves an F1 score of .83 against the human-coded gold standard. One feature of this classifier merits note: it is more likely than human coders to classify articles as not featuring Russia as a main actor. This measure is used only in supplementary analyses.

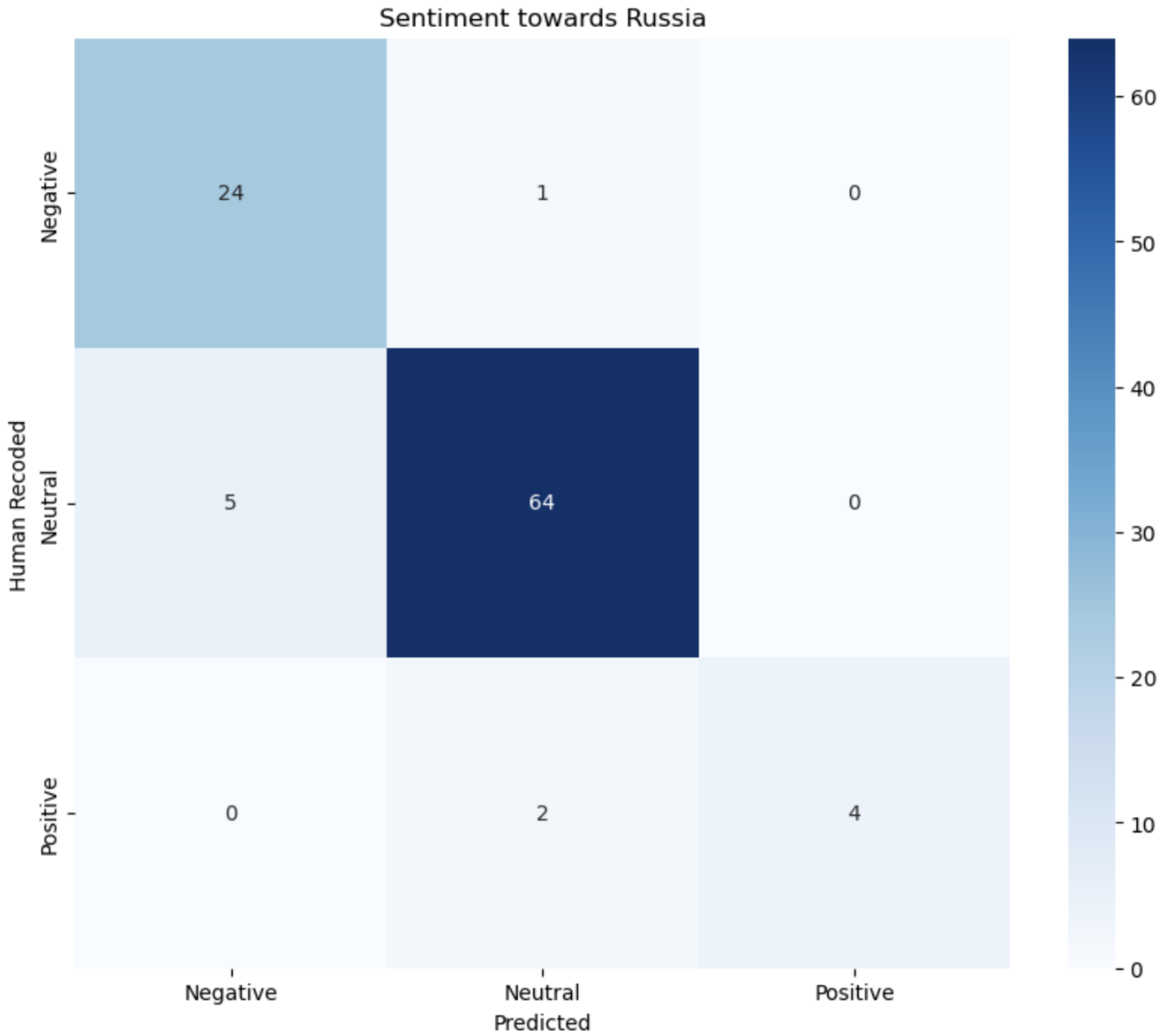

Figure A1: Confusion matrix for the GPT-4o sentiment classifier, evaluated against the human-coded gold standard.

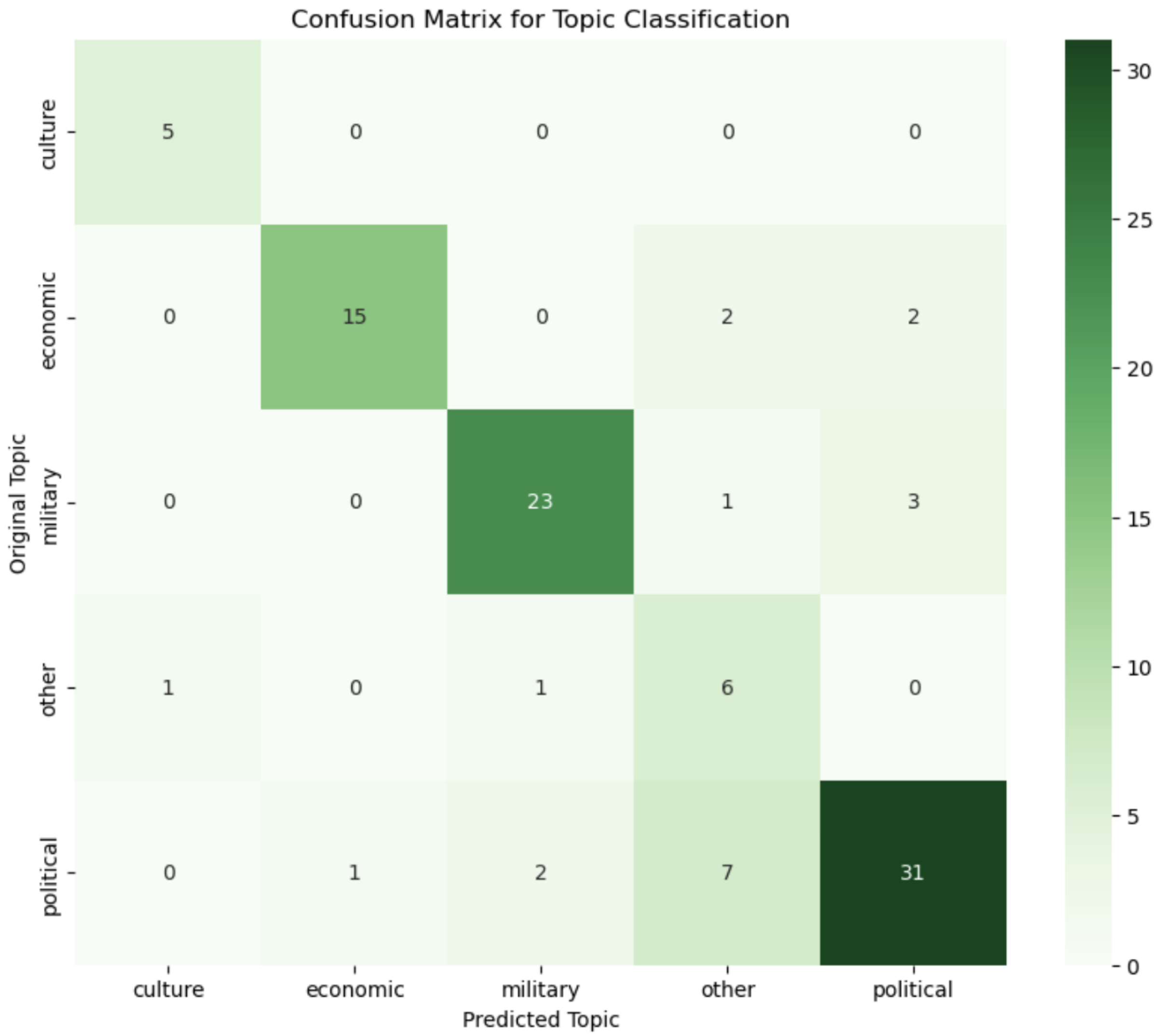

Figure A2: Confusion matrix for the GPT-4o topic classifier, evaluated against the human-coded gold standard.

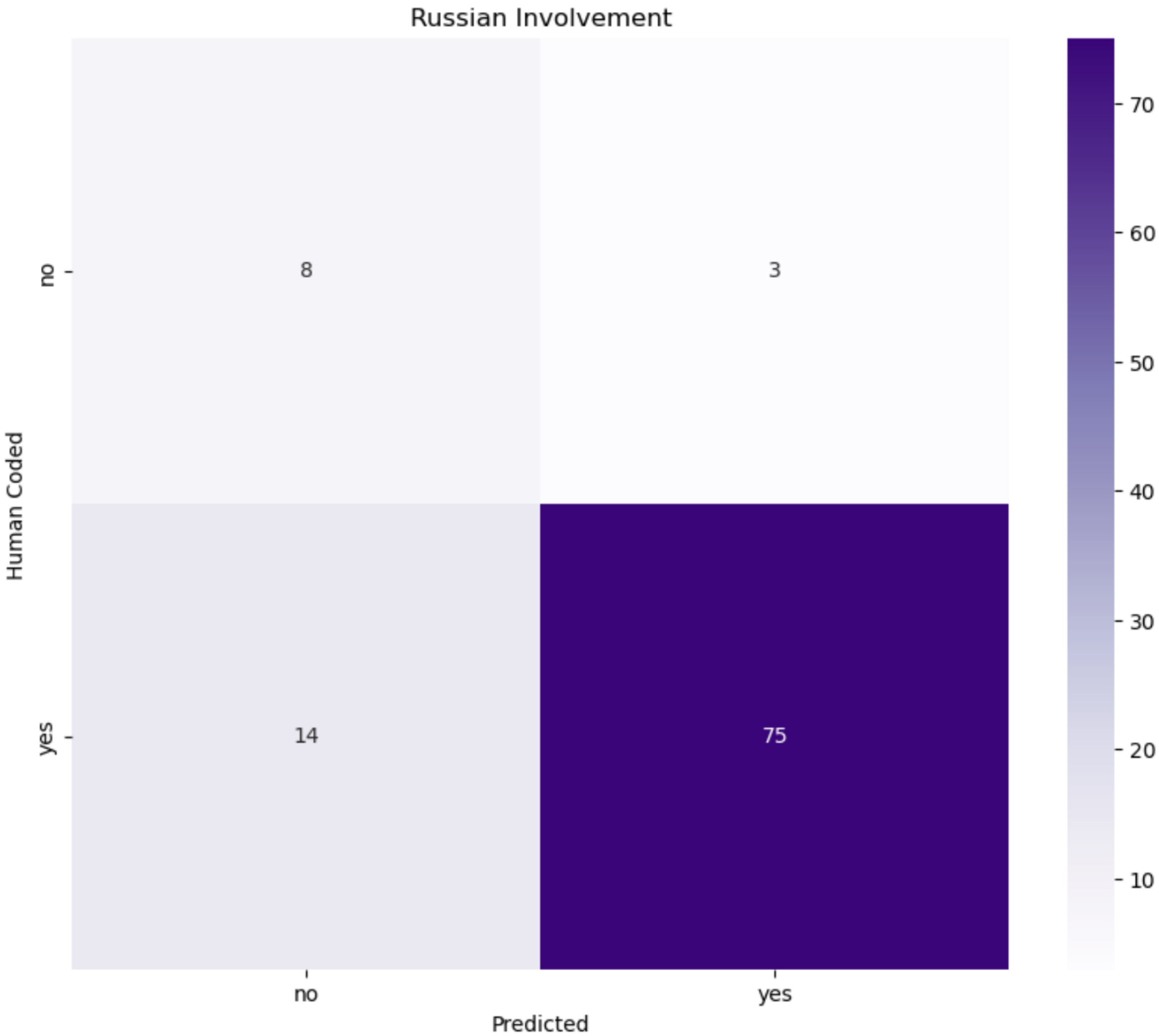

Figure A3: Confusion matrix for the GPT-4o Russian involvement classifier, evaluated against the human-coded gold standard.

Additional Descriptive Figures

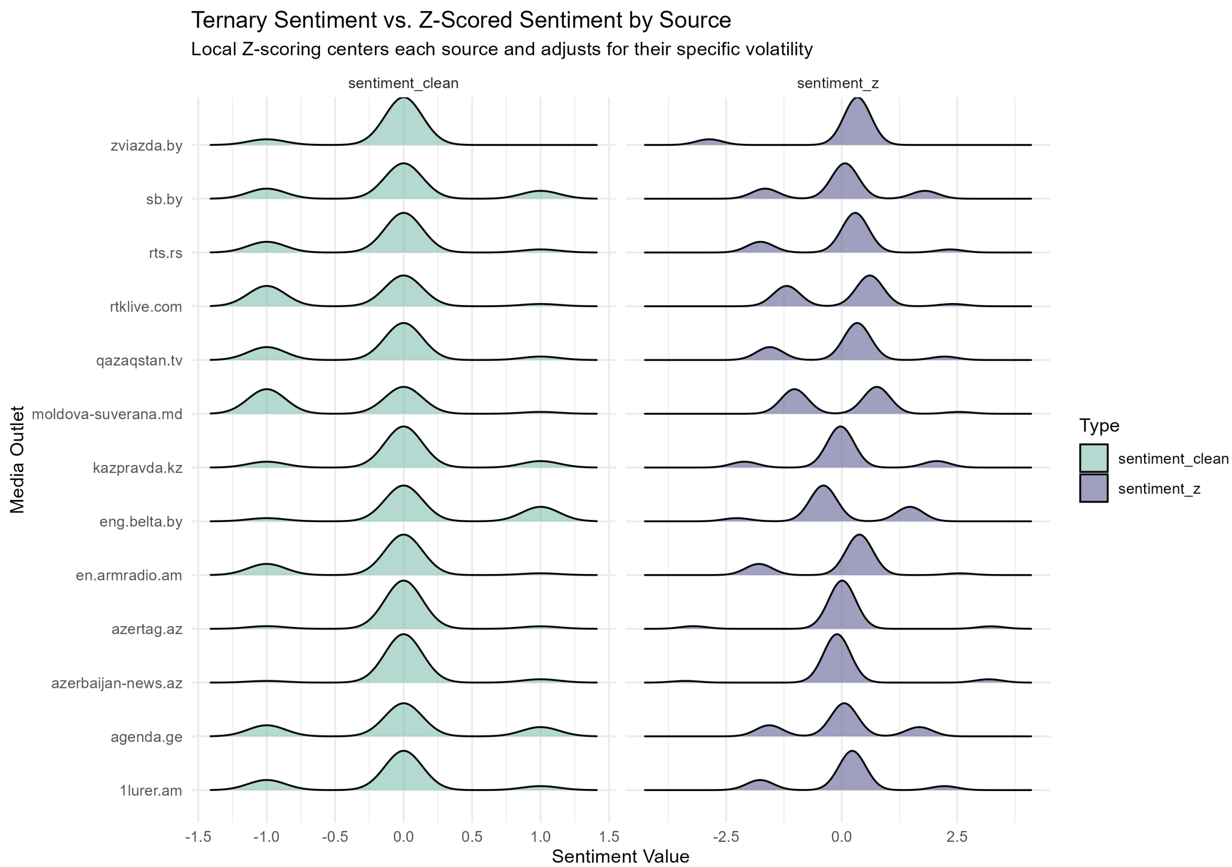

Figure A4: Comparison of raw sentiment and z-scored sentiment distributions across sources, validating the z-score standardization procedure described in the main text.

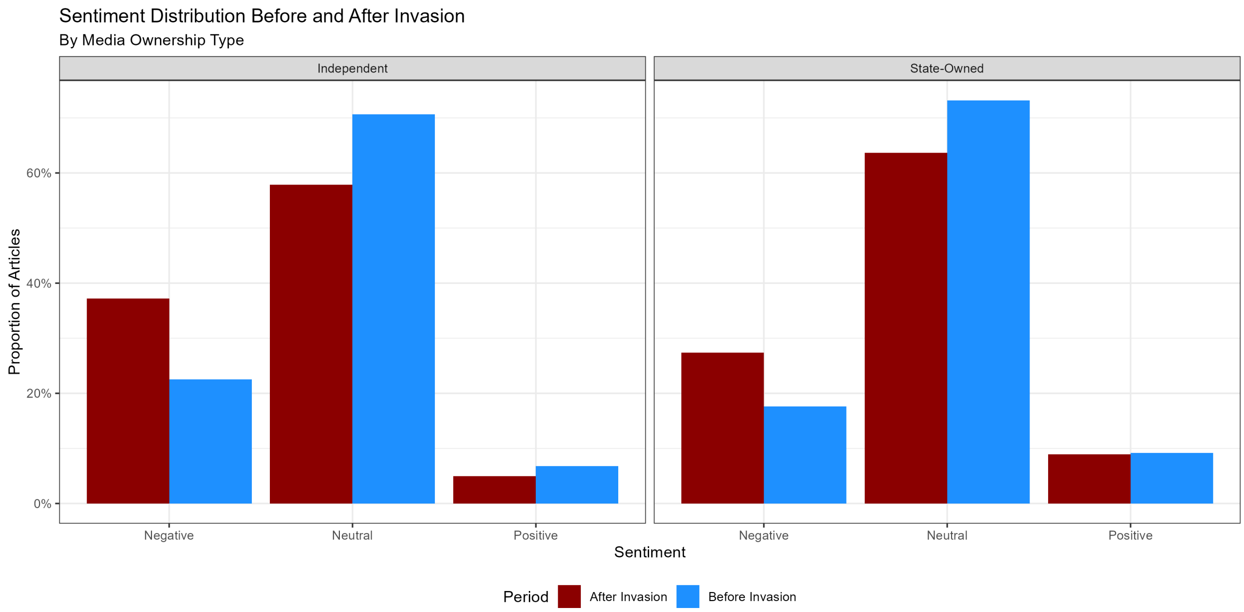

Figure A5: Distribution of sentiment scores across the sample.

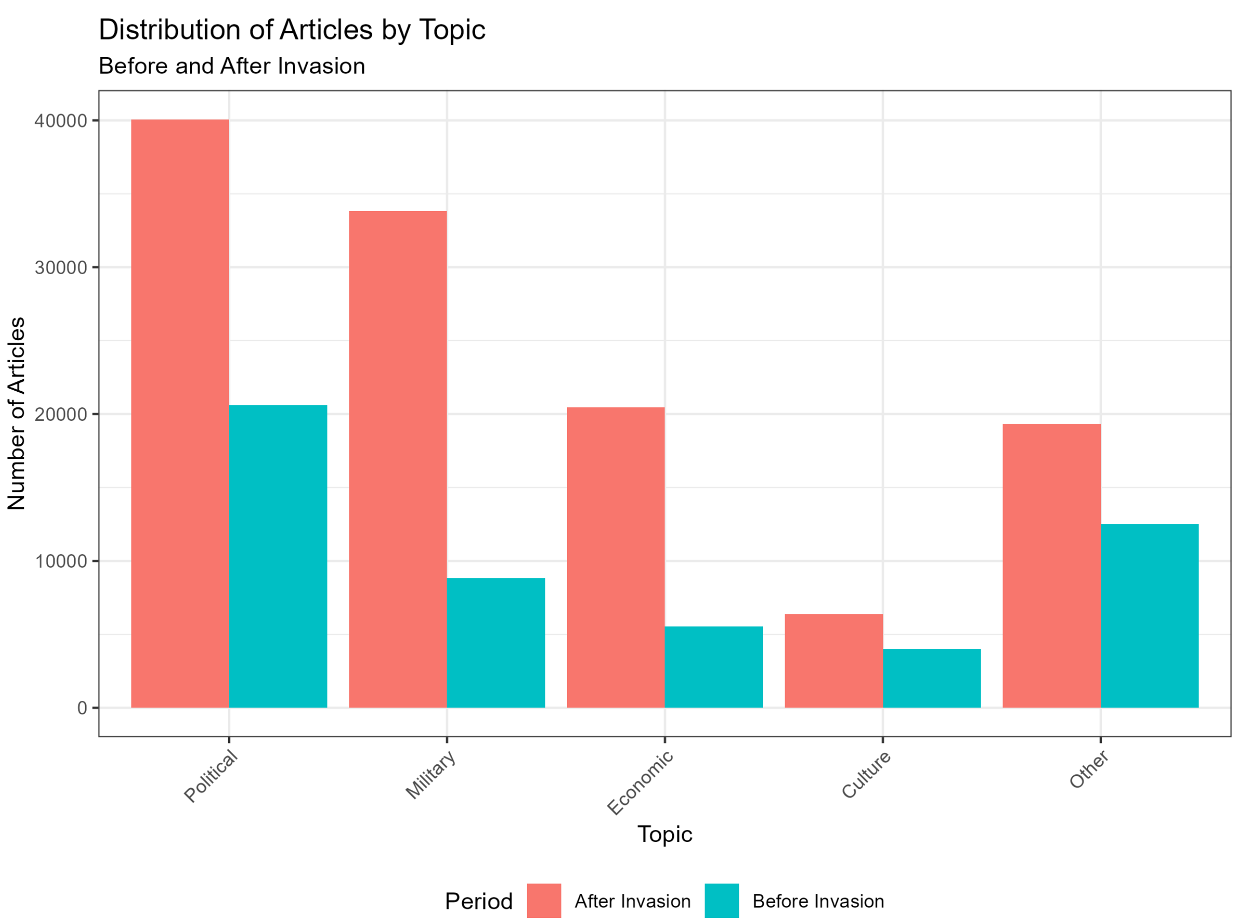

Figure A6: Distribution of articles across topic categories.

Table A1: H1 RDD — Fixed Effects Robustness (Probabilistic Time)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Invasion	-0.159***	-0.1515***	-0.1398***	-0.1398***	-0.243***	-0.091***	-0.147***
	(0.011)	(0.0092)	(0.0085)	(0.0085)	(0.016)	(0.015)	(0.013)
Topic FEs	-	-	-	X	X	X	X
State FEs	-	X	X	X	-	X	X
Source FEs	-	-	X	X	-	X	X
Source Z-Score	-	-	-	-	X	-	-
Clusters	62	62	62	62	61	—	—
Num.Obs.	174094	174094	174094	174094	174056	130525	93401
R2	0.024	0.056	0.087	0.087	0.035		
R2 Adj.	0.024	0.056	0.086	0.086	0.035		

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. Models 1-4 use the full dataset with progressively added fixed effects. Model 5 uses a Z-Score normalization with the full dataset. Models 6-7 use full fixed effects, restricting to non-military topics and non-NATO countries respectively.

H1: Additional Specifications

Table A2: H1 RDD by Topic — Z-Score (Probabilistic Time)

	Political	Military	Economic	Cultural	Other
Invasion	-0.160***	-0.422***	-0.332***	-0.569***	-0.219***
	(0.040)	(0.062)	(0.066)	(0.095)	(0.036)
Source Z-Score	X	X	X	X	X
Clusters	—	—	—	—	—
Num.Obs.	61660	43568	26230	10465	32133

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
All standard errors are clustered at the source level. All models use a source-level Z-Score normalization with the optimal bandwidth. Each column restricts to the indicated topic category.

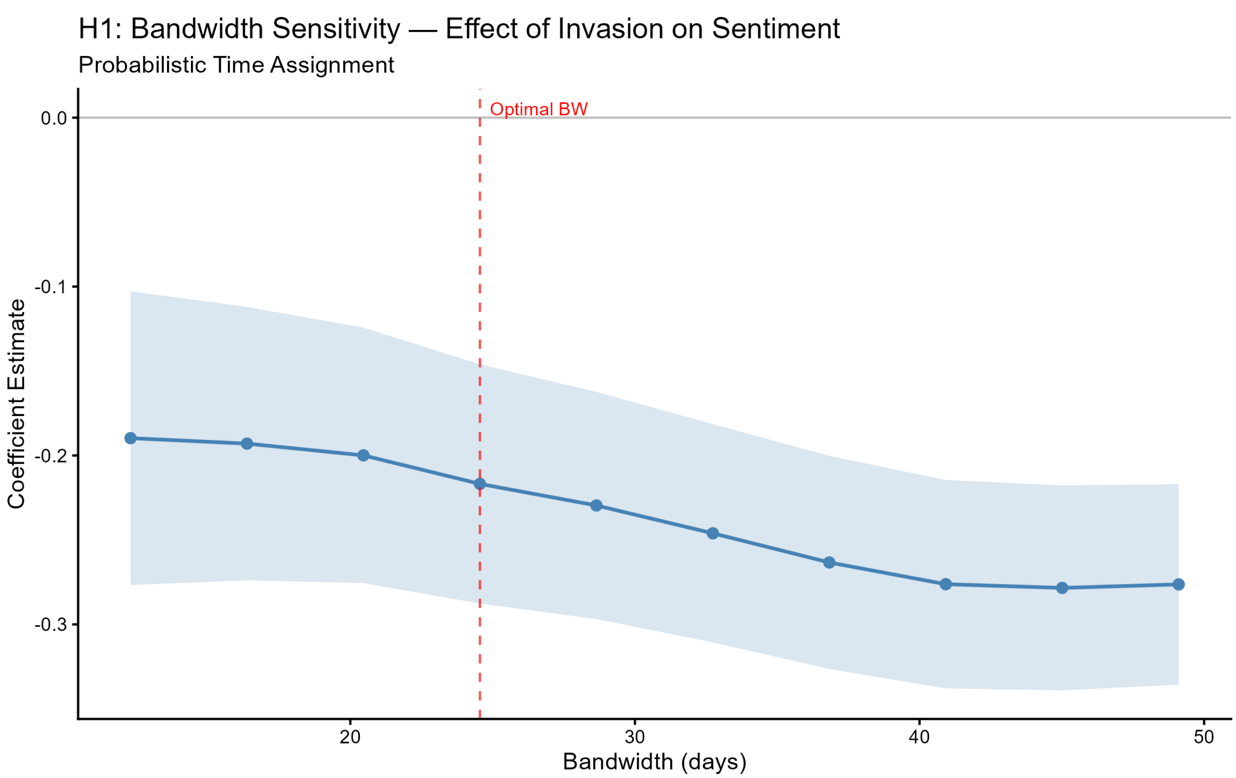

Figure A7: Bandwidth sensitivity for hypothesis 1. Each point reports the RDD coefficient estimate at the indicated bandwidth. The red dashed line marks the MSE-optimal bandwidth. Shaded region shows 95% confidence intervals.

H2: Additional Specifications

Table A3: H2 Diff-in-Disc — FE Interaction Core (Probabilistic Time)

	(1)†	(2)†	(3)†	(4)†	(5)†	(6)†	(7)†
Invasion	-0.263***	-0.16***	-0.15***	-0.125***	-0.126***	-0.122***	-0.189***
	(0.018)	(0.01)	(0.01)	(0.019)	(0.019)	(0.015)	(0.013)
State Owned x Treatment	0.090**	0.066*	0.059**	0.032	0.046	0.069+	0.146**
	(0.032)	(0.027)	(0.018)	(0.033)	(0.033)	(0.034)	(0.033)
Source Z-Score	X	-	-	-	-	-	-
State FEs	-	X	X	X	X	X	X
Source FEs	-	-	X	-	X	-	-
Topic FEs	-	X	X	X	X	X	X
Clusters	61	61	61	60	60	60	43
Num.Obs.	174056	174056	174056	37622	37622	27315	32095
R2	0.036	0.061	0.087	0.065	0.084	0.042	0.045
R2 Adj.	0.036	0.061	0.087	0.065	0.082	0.042	0.045

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
All standard errors are clustered at the source level. Model 1 uses a source-level Z-Score normalization with the full dataset. Model 2 uses the full dataset with state + topic FEs. Model 3 adds source FEs. Model 4 uses the optimal bandwidth. Model 5 uses optimal bandwidth with full FEs. Model 6 omits military topics. Model 7 omits NATO allies. Model 8 restricts to Russian-related articles. Model 9 uses matched data. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=4999, null imposed), clustered by source domain.

Table A4: H2 Diff-in-Disc — Full FE + DID Robustness (Probabilistic Time)

	(1)†	(2)†	(3)†
Invasion	-0.124***	-0.217***	-0.178***
	(0.016)	(0.017)	(0.009)
State Owned x Treatment	0.078+	0.138*	0.074**
	(0.033)	(0.038)	(0.027)
State FEs	X	X	X
Source FEs	X	X	X
Topic FEs	X	X	X
Clusters	60	42	60
Num.Obs.	27315	20604	37622
R2	0.064	0.092	0.057
R2 Adj.	0.062	0.090	0.057

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
All standard errors are clustered at the source level. Model 1 uses non-military subset with full FEs. Model 2 uses Russian-related articles with full FEs. Model 3 is a DID specification within the optimal bandwidth. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=4999, null imposed), clustered by source domain.

Table A5: H2 Diff-in-Disc by Topic — Full Data (Probabilistic Time)

	Political	Military	Economic	Cultural	Other
Invasion	-0.175***	-0.316***	-0.31***	-0.44***	-0.216***
	(0.027)	(0.029)	(0.06)	(0.05)	(0.042)
State Owned x Treatment	0.076+	0.022	0.086	0.02	0.220**
	(0.041)	(0.046)	(0.096)	(0.13)	(0.072)
State FEs	X	X	X	X	X
Source FEs	-	-	-	-	-
Topic FEs	-	-	-	-	-
Clusters	61	61	61	61	61
Num.Obs.	61660	43568	26230	10465	32133
R2	0.006	0.019	0.028	0.027	0.009
R2 Adj.	0.006	0.019	0.028	0.026	0.009

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. All models use the optimal bandwidth with state fixed effects. Each column restricts to the indicated topic category. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=4999, null imposed), clustered by source domain.

Table A6: H2 Leave-One-Out Country Robustness (Probabilistic Time)

Country Dropped	Coefficient	Std. Error	p-value	N	Bandwidth
Standard Model					
Albania	0.171**	(0.060)	0.006	27,709	21.5
Azerbaijan	0.133*	(0.064)	0.044	24,124	18.9
Armenia	0.157*	(0.065)	0.020	23,875	19.4
Belarus	0.152*	(0.066)	0.026	24,849	19.2
Hungary	0.156**	(0.054)	0.005	29,259	24.5
Kazakhstan	0.169**	(0.060)	0.007	30,457	22.8
Kosovo	0.159*	(0.062)	0.013	30,302	27.3
Moldova	0.134*	(0.062)	0.035	29,167	22.0
Serbia	0.121	(0.066)	0.073	25,116	19.7
Turkey	0.210**	(0.074)	0.006	29,326	28.4
Georgia	0.131*	(0.059)	0.031	28,633	21.4
Matched Model					
Albania	0.180**	(0.058)	0.003	27,704	21.5
Azerbaijan	0.147*	(0.064)	0.027	24,121	18.9
Armenia	0.170**	(0.063)	0.009	23,871	19.4
Belarus	0.162*	(0.065)	0.016	24,764	19.2
Hungary	0.165**	(0.054)	0.004	29,248	24.5
Kazakhstan	0.181**	(0.060)	0.004	30,452	22.8
Kosovo	0.170**	(0.064)	0.010	30,292	27.3
Moldova	0.144*	(0.061)	0.022	29,162	22.0
Serbia	0.141*	(0.065)	0.034	25,112	19.7
Turkey	0.216**	(0.075)	0.006	29,322	28.4
Georgia	0.144*	(0.058)	0.017	28,628	21.4

Leave-one-out robustness analysis. Each row shows results when excluding the specified country. Bandwidth is recalculated for each subset. Coefficient shown is for the treatment:state_owned interaction term. Standard errors clustered at source domain level.

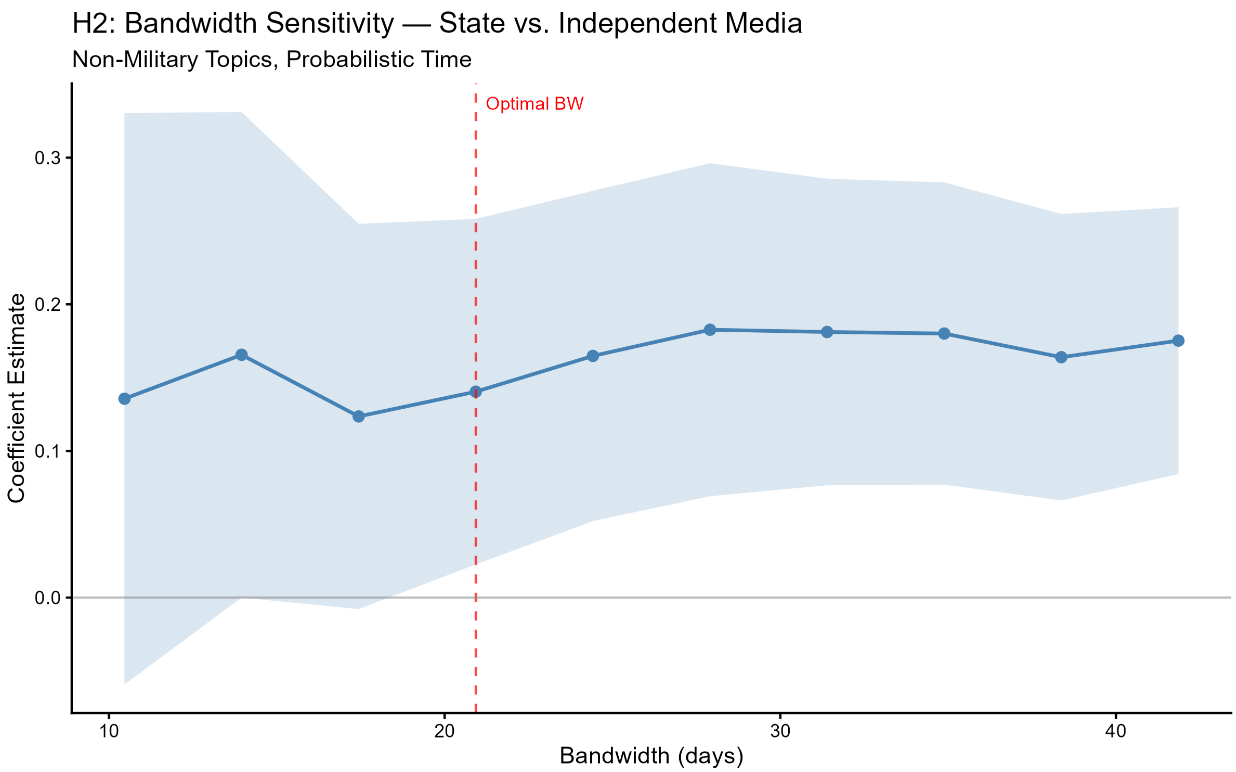

Figure A8: Bandwidth sensitivity for hypothesis 2. Each point reports the RDID interaction coefficient estimate at the indicated bandwidth. The red dashed line marks the MSE-optimal bandwidth. Shaded region shows 95% confidence intervals.

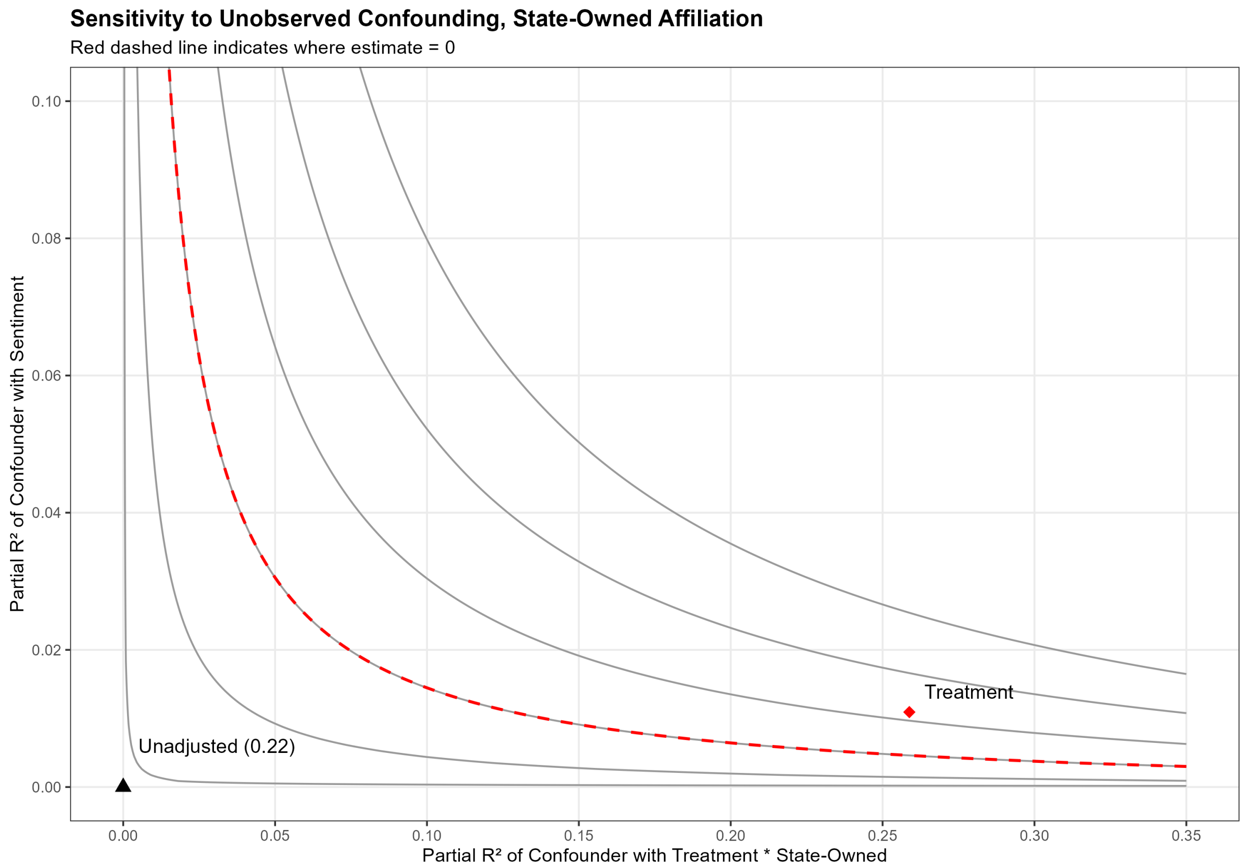

Figure A9: Sensitivity of the state-owned $\times$ treatment interaction to unobserved confounding, following Cinelli and Hazlett (2020). The red dashed line indicates the combinations of confounder strength that would reduce the estimate to zero.

H3: Additional Specifications

Table A7: H3a Alignment — FE Interaction + Robustness (Probabilistic Time)

	(1)†	(2)†	(3)†	(4)†	(5)†	(6)†	(7)†	(8)†	(9)†
Invasion	-0.150	-0.081	-0.018	0.079	0.061	-0.042	0.098	0.027	0.029
	(0.094)	(0.055)	(0.037)	(0.050)	(0.051)	(0.047)	(0.069)	(0.034)	(0.046)
CSTO x Treatment	-0.021	-0.0025	-0.107	-0.202*	-0.222*	-0.074	-0.38*	-0.149+	-0.179
	(0.102)	(0.0579)	(0.052)	(0.061)	(0.069)	(0.058)	(0.11)	(0.049)	(0.063)
Source Z-Score	X	-	-	-	-	-	-	-	-
State FEs	-	X	X	X	X	X	X	X	X
Source FEs	-	-	-	-	-	X	-	X	X
Topic FEs	-	X	X	X	X	X	X	X	X
Matched	-	-	-	-	-	-	X	-	-
Clusters	13	13	13	13	13	13	13	13	13
Num.Obs.	26911	26911	7008	5917	4651	7008	5721	5917	4651
R2	0.023	0.086	0.059	0.049	0.076	0.094	0.014	0.088	0.120
R2 Adj.	0.023	0.085	0.057	0.047	0.074	0.091	0.013	0.085	0.116

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. Sample restricted to state-owned media in non-NATO countries. Model 1 uses a source-level Z-Score normalization with the full dataset. Model 2 uses the full dataset with FEs. Model 3 uses the optimal bandwidth. Model 4 omits military topics. Model 5 restricts to Russian-related articles. Model 6 uses optimal bandwidth with full FEs. Model 7 uses matched data. Models 8-9 add full FEs to non-military and Russian subsets. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=4999, null imposed), clustered by source domain.

Table A8: H3a Alignment by Topic — Z-Score (Probabilistic Time)

	Political	Military	Economic	Cultural	Other
Invasion	0.0051	-0.46+	0.15	-1.35**	0.33
	(0.0725)	(0.25)	(0.11)	(0.34)	(0.60)
CSTO x Treatment	-0.26	0.29	-0.39	0.74	-0.32
	(0.12)	(0.28)	(0.21)	(0.45)	(0.60)
Source Z-Score	X	X	X	X	X
Clusters	13	13	13	13	13
Num.Obs.	2661	1783	1343	501	2206
R2	0.004	0.038	0.021	0.099	0.028
R2 Adj.	0.001	0.034	0.016	0.086	0.025

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All models use a source-level Z-Score normalization with the optimal bandwidth. Sample restricted to state-owned media in non-NATO countries. Each column restricts to the indicated topic category. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=4999, null imposed), clustered by source domain. ‡ Cluster-robust SEs from rdrobust.

Table A9: H3a Alignment by Topic — Full Data (Probabilistic Time)

	Political	Military	Economic	Cultural	Other
Invasion	-0.085	-0.33*	-0.052	-0.86*	0.12
	(0.068)	(0.12)	(0.154)	(0.33)	(0.39)
CSTO x Treatment	-0.050	0.11	-0.29	0.60	-0.11
	(0.098)	(0.15)	(0.17)	(0.34)	(0.39)
State FEs	X	X	X	X	X
Source FEs	-	-	-	-	-
Topic FEs	-	-	-	-	-
Clusters	13	13	13	13	13
Num.Obs.	9049	4376	4884	2401	6201
R2	0.011	0.021	0.018	0.020	0.018
R2 Adj.	0.011	0.019	0.016	0.017	0.017

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. Sample restricted to state-owned media in non-NATO countries. All models use the optimal bandwidth with state fixed effects. Each column restricts to the indicated topic category. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=4999, null imposed), clustered by source domain.

Table A10: H3a Leave-One-Out Country Robustness (Probabilistic Time)

Country Dropped	Coefficient	Std. Error	p-value	N	Bandwidth
Standard Model					
Belarus	-0.188	(0.136)	0.201	5,664	75.2
Armenia	-0.258*	(0.102)	0.030	4,884	37.2
Georgia	-0.230*	(0.101)	0.043	5,830	41.3
Azerbaijan	-0.361*	(0.136)	0.024	5,232	39.6
Kazakhstan	-0.249*	(0.085)	0.015	5,286	36.3
Moldova	-0.326*	(0.114)	0.015	5,706	39.0
Kosovo	-0.298	(0.138)	0.054	5,046	37.1
Serbia	-0.387**	(0.112)	0.005	5,715	40.2
Matched Model					
Belarus	-0.203	(0.142)	0.186	5,314	75.2
Armenia	-0.344*	(0.109)	0.010	4,589	37.2
Georgia	-0.280*	(0.107)	0.024	5,374	41.3
Azerbaijan	-0.387*	(0.136)	0.017	4,946	39.6
Kazakhstan	-0.277**	(0.076)	0.004	4,971	36.3
Moldova	-0.383**	(0.110)	0.005	5,288	39.0
Kosovo	-0.348*	(0.130)	0.021	4,505	37.1
Serbia	-0.407**	(0.112)	0.004	5,103	40.2

Leave-one-out robustness analysis. Each row shows results when excluding the specified country. Bandwidth is recalculated for each subset. Coefficient shown is for the treatment:csto interaction term. Standard errors clustered at source domain level.

Table A11: Treatment Effect Decomposition by Source

	Estimate	SE (Pairs)	p (Pairs)	SE (Bayes)	p (Bayes)	N
Panel A: Group-Level Effects						
CSTO	-0.211	(0.071)	0.000***	(0.058)	0.000***	9
Non-CSTO	-0.004	(0.134)	0.973	(0.102)	0.988	4
Difference	-0.206	(0.152)	0.230	(0.117)	0.116	13
Panel B: Source-Level Effects						
1lurer.am (CSTO)	-0.354	—	—	—	—	641
azerbaijan-news.az (CSTO)	-0.147	—	—	—	—	125
azertag.az (CSTO)	-0.057	—	—	—	—	625
en.armradio.am (CSTO)	-0.276	—	—	—	—	260
eng.belta.by (CSTO)	-0.129	—	—	—	—	432
kazpravda.kz (CSTO)	-0.853	—	—	—	—	241
qazaqstan.tv (CSTO)	-0.349	—	—	—	—	57
sb.by (CSTO)	-0.156	—	—	—	—	2031
zviazda.by (CSTO)	0.281	—	—	—	—	41
agenda.ge (Neutral)	0.150	—	—	—	—	375
moldova-suverana.md (Neutral)	-0.221	—	—	—	—	215
rtklive.com (Neutral)	0.144	—	—	—	—	732
rts.rs (Neutral)	-0.387	—	—	—	—	315

Panel A reports N-weighted average treatment effects by CSTO group, estimated from separate RDD regressions within each source-topic cell. Pairs bootstrap resamples sources with replacement (B = 9,920). Bayesian bootstrap draws Dirichlet(1,...,1) source weights (Rubin 1981; B = 9,920). Panel B reports source-level treatment effects (N-weighted average across topics within each source). N in Panel A is number of sources; N in Panel B is articles per source. Sample restricted to state-owned, non-NATO, non-military articles within the optimal bandwidth. Source-level Z-score normalization applied.

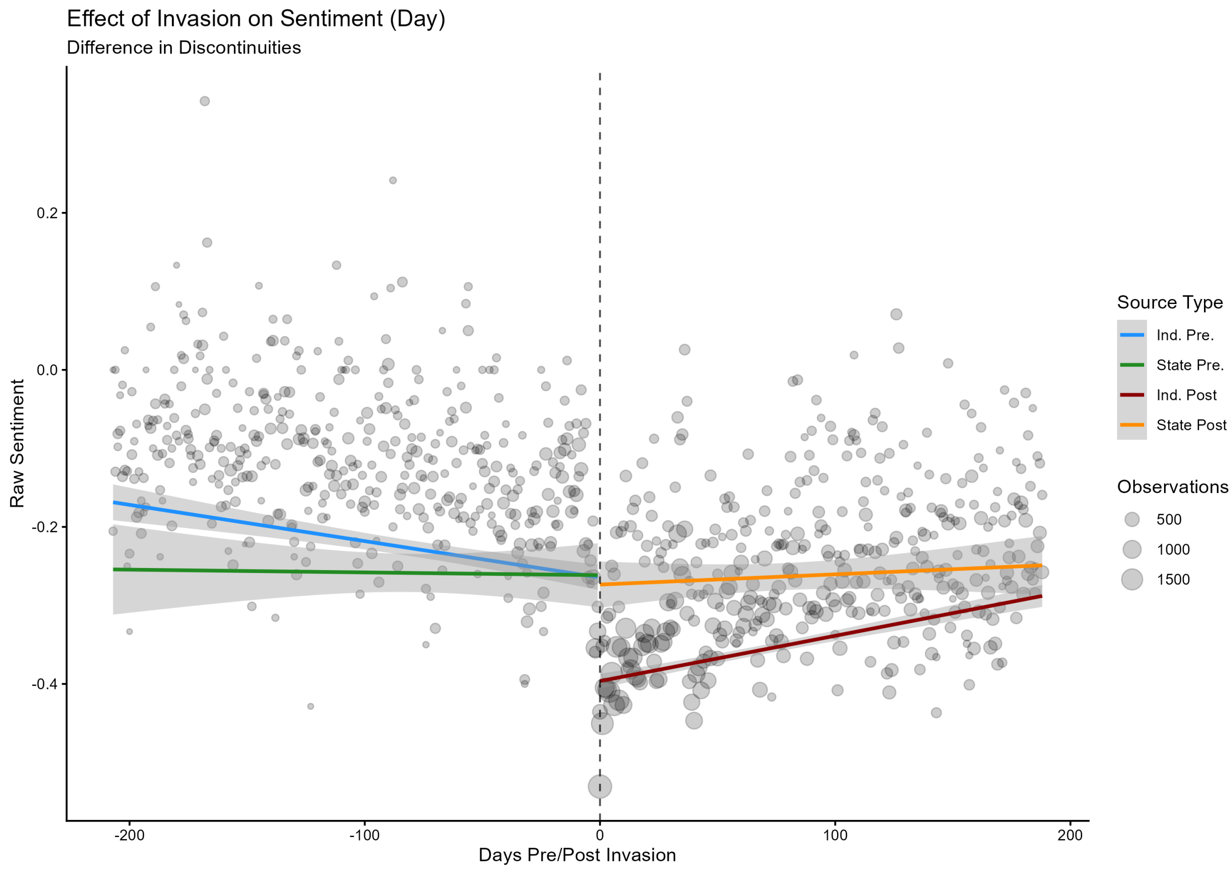

Figure A10: RDD estimates of the effect of the invasion on sentiment toward Russia in neutral countries, by media type. Pre-invasion state media sentiment sits well above the scale floor and post-invasion independent media falls significantly below state media, confirming that the null H3 result for neutral state media does not reflect a measurement constraint.

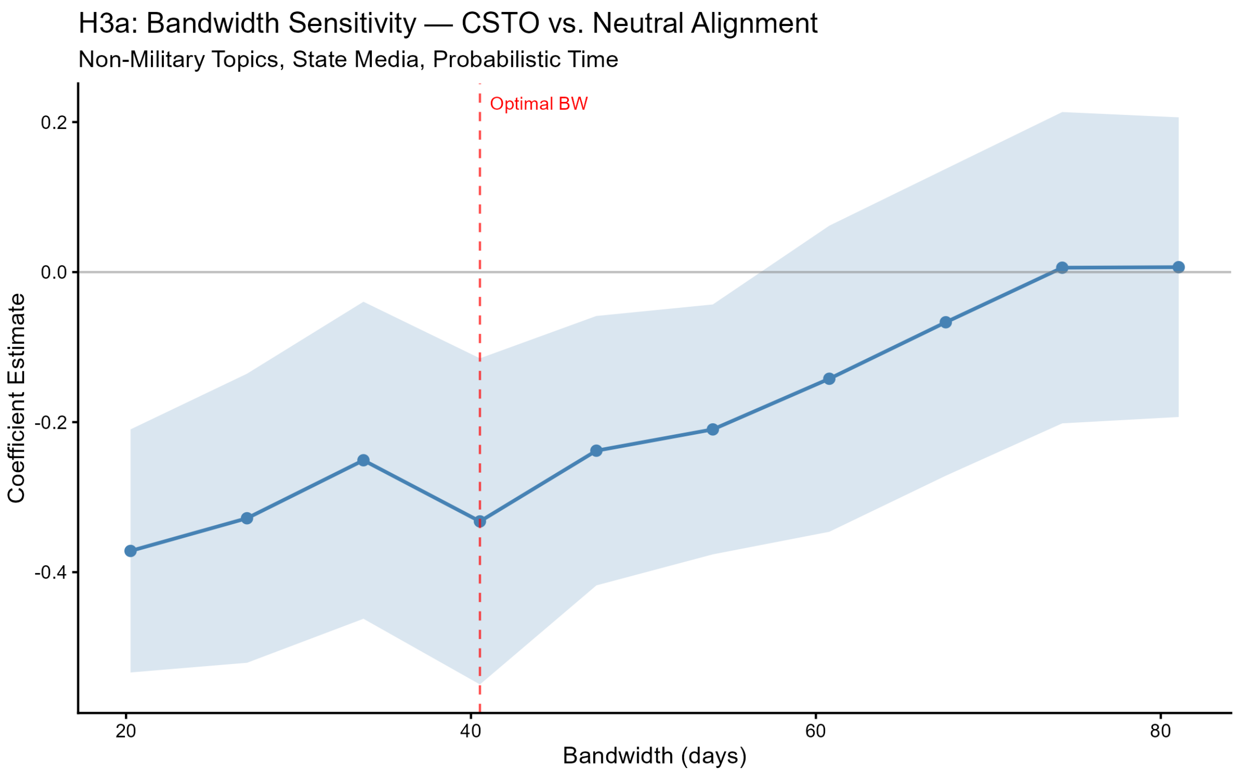

Figure A11: Bandwidth sensitivity for hypothesis 3. Each point reports the RDID interaction coefficient estimate at the indicated bandwidth. The red dashed line marks the MSE-optimal bandwidth. Shaded region shows 95% confidence intervals.

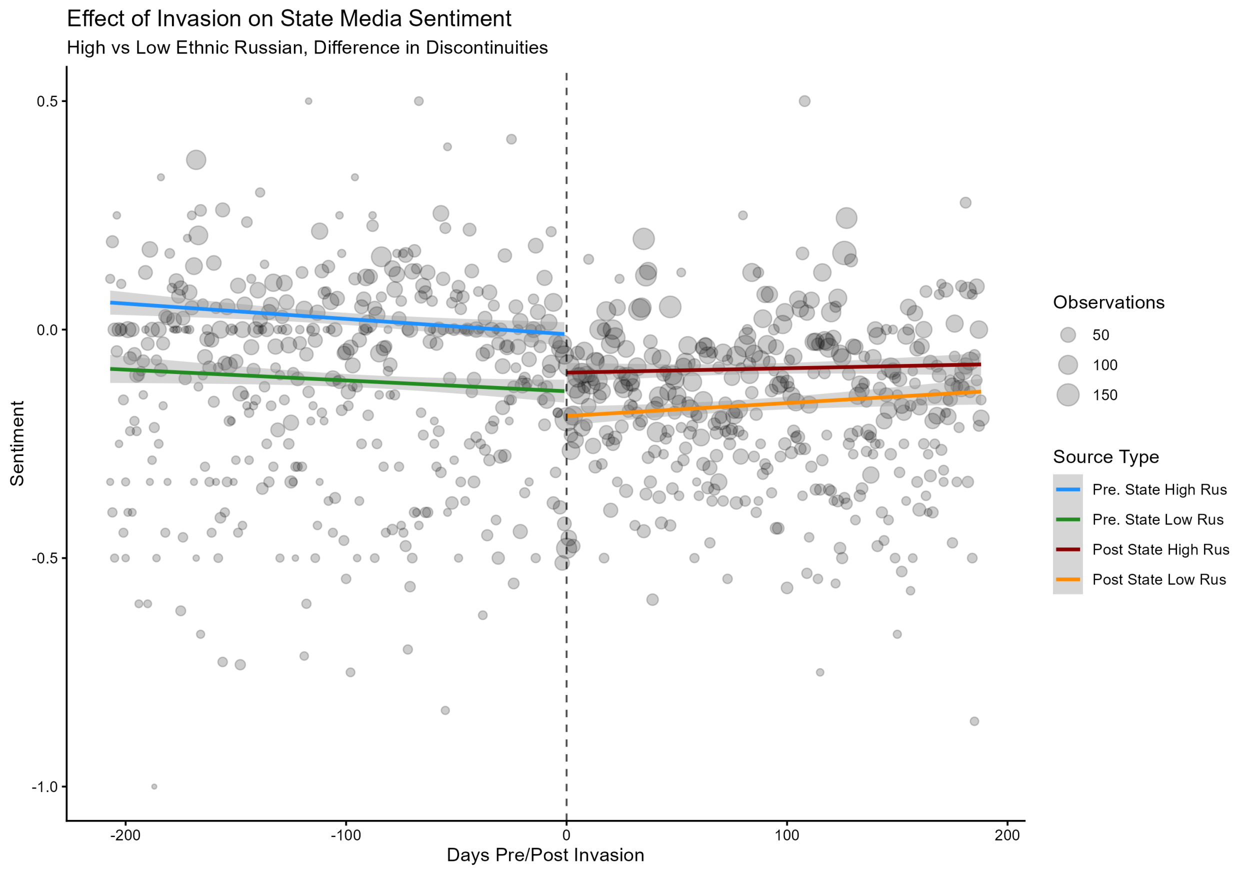

Figure A12: RDD estimates of the effect of the invasion on sentiment toward Russia in state-owned media, by ethnic Russian population share. Sample restricted to state-owned media in non-NATO countries. Each point represents one day, sized by the number of articles published. The dashed line marks February 24$^{\text{th}}$, 2022.

Placebo Test

Table A12: Placebo Test Summary (Probabilistic Time)
December 7, 2021 False Treatment Date

Model	Estimate	Std. Error	p-value	Bandwidth	N
(1) H1: Full Model (RDD)	0.016	(0.033)	0.637	771160.0	3,492
(2) H1: Z-Score (RDD)	0.000	(0.063)	0.994	901739.1	4,865
(3) H2: Diff-in-Disc	0.250	(0.119)	0.088	8.9	4,691
(4) H3: Non-Military	0.705	(0.287)	0.140	21.3	1,983

Placebo tests using December 7, 2021 (Biden's announcement) as a false treatment date. Data restricted to pre-invasion period (before Feb 19, 2022). Model 1 uses rdrobust; remaining models use felm with optimal bandwidth. Non-significant treatment effects suggest parallel trends assumption holds.